

Microsoft Power BI Dashboards

Step

by

Step

Errin O'Connor



PRACTICE FILES



Microsoft Power BI Dashboards Step by Step

By Errin O'Connor

Microsoft Power BI Dashboards Step by Step

Published with the authorization of Microsoft Corporation by:
Pearson Education, Inc.

Copyright © 2019 by Pearson Education, Inc.

All rights reserved. Printed in the United States of America. This publication is protected by copyright, and permission must be obtained from the publisher prior to any prohibited reproduction, storage in a retrieval system, or transmission in any form or by any means, electronic, mechanical, photocopying, recording, or likewise. For information regarding permissions, request forms, and the appropriate contacts within the Pearson Education Global Rights & Permissions Department, please visit www.pearsoned.com/permissions/. No patent liability is assumed with respect to the use of the information contained herein. Although every precaution has been taken in the preparation of this book, the publisher and author assume no responsibility for errors or omissions. Nor is any liability assumed for damages resulting from the use of the information contained herein.

ISBN-13: 978-1-509-30803-3

ISBN-10: 1-509-30803-2

Library of Congress Control Number: 2018957364

1 18

Trademarks

Microsoft and the trademarks listed at <http://www.microsoft.com> on the “Trademarks” webpage are trademarks of the Microsoft group of companies. All other marks are property of their respective owners.

Warning and Disclaimer

Every effort has been made to make this book as complete and as accurate as possible, but no warranty or fitness is implied. The information provided is on an “as is” basis. The author, the publisher, and Microsoft Corporation shall have neither liability nor responsibility to any person or entity with respect to any loss or damages arising from the information contained in this book.

Special Sales

For information about buying this title in bulk quantities, or for special sales opportunities (which may include electronic versions; custom cover designs; and content particular to your business, training goals, marketing focus, or branding interests), please contact our corporate sales department at corpsales@pearsoned.com or (800) 382-3419.

For government sales inquiries, please contact governmentsales@pearsoned.com. For questions about sales outside the U.S., please contact intlcs@pearson.com.

Editor-in-Chief

Brett Bartow

Acquisitions Editor

Trina MacDonald

Development Editor

Songlin Qiu

Managing Editor

Sandra Schroeder

Senior Project Editor

Tracey Croom

Copy Editor

Kate Shoup

Production Editor

Dan Foster

Indexer

Valerie Haynes Perry

Proofreader

Dan Foster

Technical Editor

Pierstefano Tucci

Editorial Assistant

Cindy Teeters

Cover Designer

Twist Creative, Seattle

Compositor

Danielle Foster

Graphics

Vived Graphics

*I would like to dedicate this book to my wife, Linda,
and my two boys, Carter and Owen.
You guys are my whole world,
and I couldn't have done this without your support.*

This page intentionally left blank

Contents

Acknowledgments	x
About the author.....	xi



Introduction	xiii
Who this book is for	xiii
The Step by Step approach.....	xiii
Features and conventions.....	xiv
Download the practice files	xvi
Ebook edition.....	xvii
Get support and give feedback.....	xvii

1

Introduction to Power BI.....	1
Sidebar: Practice files	2
An introduction to Power BI.....	2
Sidebar: Real-world scenario	3
Sidebar: Using Power BI effectively to streamline organizational growth.....	4
Sidebar: A word on datasets	5
Power BI features.....	5
Power BI platforms	8
Power BI licensing models	10
Set up Power BI	12
Prepare a dashboard in Power BI service.....	18
General note for Power BI security.....	22
Skills review.....	22
Practice tasks	23

2

Power BI architecture and administration	25
Power BI architecture.....	26
The Power BI Admin Portal.....	29
Power BI data models and report development	44
Sidebar: Relationship view	51
Skills review.....	55
Practice tasks	56

3

Visuals in Power BI	59
Overview of visuals in Power BI.....	60
Power BI visuals out of the box	60
Third-party visuals in Power BI	69
Sidebar: Other ways to import third-party visuals into Power BI	71
Sidebar: More options.....	72
Drill through Power BI visuals	72
Apply filters in Power BI reports.....	73
Sidebar: Slicers	75
Skills review.....	75
Practice tasks	76

4

Power BI report development	79
Sidebar: Practice files	80
Data sources supported by Power BI.....	80
Build a Power BI report in Power BI Desktop	85
Sidebar: Relationship view	89
Publish your report to Power BI service	97
Skills review.....	99
Practice tasks	100

5

Use DAX in Power BI	101
Calculated columns and measures	102
Sidebar: DAX capabilities	109
DAX expressions syntax and context	109
DAX functions	111
Skills review	124
Practice tasks	125

6

Develop Power BI reports from Excel	127
Sidebar: Practice files	128
Import data from an Excel workbook into Power BI	128
Add a measure and a calculated column	129
Add Power BI visuals to the report	131
Arrange the visuals	136
Prepare the report for mobile view	137
Publish and view the report	138
Prepare a dashboard from the report	139
Skills review	140
Practice tasks	141

7

Develop Power BI reports from SharePoint Online	143
Sidebar: Practice files	144
Build a Power BI report based on data from a SharePoint Online list	144
Build a Power BI report based on data from a SharePoint Online folder	154
Refresh report data in a Power BI report	161
Embed Power BI reports in SharePoint Online	164
Skills review	166
Practice tasks	167

8

Develop Power BI reports from SharePoint On-Premises169

Sidebar: Practice files	170
Import data from a SharePoint On-Premises list into Power BI	170
Create a Power BI report	172
Set up a data gateway	173
Configure Power BI to use the data gateway and set up refresh functionality	180
Skills review	184
Practice tasks	185

9

Develop Power BI reports from SQL.....187

Sidebar: Practice files	188
Import data from a SQL Server database into Power BI	188
Create a Power BI report	193
Set up a data gateway	195
Configure Row-Level Security (RLS) in Power BI	198
Sidebar: Limitations of RLS in Power BI	203
Skills review	203
Practice tasks	204

10

Develop Power BI reports from SSAS cube 207

Sidebar: Practice files	208
Basic terminology	208
Import data from an SSAS cube into Power BI	209
Create a Power BI report	214
Set up and configure a data gateway	216
Skills review	219
Practice tasks	220

11

Develop Power BI reports from Azure SQL	223
Sidebar: Practice files	224
Overview of the Azure SQL portal	224
Import data from an Azure SQL database into Power BI.....	226
Create a Power BI report.....	232
Set up a live data connection and data refresh functionality.....	234
Skills review.....	238
Practice tasks	239

12

Develop Power BI reports from Oracle	241
Sidebar: Practice files	242
Import data from an Oracle database into Power BI	242
Create a Power BI report.....	248
Set up and configure a data gateway	251
Skills review.....	256
Practice tasks	257

13

Develop Power BI reports from Dynamics 365	259
Sidebar: Practice files	260
Import data from Dynamics 365 into Power BI Desktop.....	260
Sidebar: URL syntax	263
Create a Power BI report.....	264
Import data from Dynamics 365 into Power BI service	266
Embed a Power BI dashboard into a Dynamics 365 dashboard	269
Set up data refresh functionality	272
Skills review.....	275
Practice tasks	276
 Index.....	 279

Acknowledgments

I would like to thank my wonderful wife, Linda, and my two amazing boys, Carter and Owen, for all their support. This book would not have been possible without you.

I appreciate the time and efforts of Tracey Croom, Kate Shoup, and the entire team at Microsoft Press—past and present—who made this and so many other books possible.

I would also like to thank my team at EPC Group for all their support and their contributions.

About the Author

Errin O'Connor is the founder and chief architect of EPC Group. O'Connor focuses his efforts on implementing technologies in organizations throughout the country and around the globe.

For more than 20 years, O'Connor has assisted in managing EPC Group's corporate strategy and architects the proven methodologies around business intelligence, collaboration, enterprise content management, and custom application development that have set EPC Group apart from its competitors.

EPC Group has completed more than 5,000 Power BI, SharePoint, Office 365, and Microsoft stack-related implementations, including efforts around business intelligence (BI), custom application development, hybrid cloud strategy, Microsoft Azure, Office 365, SQL Server, Microsoft Intune, Amazon Web Services (AWS), and Microsoft Project Server. Founded in 1997, EPC Group has pioneered the way organizations implement BI platforms and collaborate.

In addition to this book, O'Connor is also the author or co-author of three other books: *SharePoint 2013 Field Guide: Advice from the Consulting Trenches* (Sams Publishing, 2014), *Microsoft SharePoint Foundation 2010: Inside Out* (Microsoft Press, 2011), and *Windows SharePoint Services 3.0: Inside Out* (Microsoft Press, 2007).

O'Connor is a frequent speaker at Microsoft Power BI, Office 365, and Microsoft SharePoint events throughout the United States and Canada.

For more information about O'Connor and EPC Group, visit www.epcgroup.net or contact the company at contact@epcgroup.net.

This page intentionally left blank

Introduction



Welcome! This *Step by Step* book has been designed so you can read it from the beginning to learn about Microsoft Power BI and then build your skills as you learn to perform increasingly specialized procedures. If you prefer, you can jump in wherever you need guidance to perform Power BI–related tasks. The how-to-steps are delivered in a concise manner. You will find full-color graphics that support the instructional content in this book.

Who this book is for

Microsoft Power BI Dashboards Step by Step is designed for use as a learning and reference resource by home and business users who want to create world-class business analysis dashboards that integrate today's most widely used data sources. The content of this book is designed to be useful for people who have previously used Power BI as well as for people who are discovering Power BI for the first time. This full-color, hands-on guide walks you through the key decisions analysts and developers need to make up front, and introduces all the concepts, skills, and techniques you'll need to achieve your goals.

The Step by Step approach

The book's coverage is divided into several parts representing general Power BI skill sets. Each part is divided into chapters representing skill set areas, and each chapter is divided into topics that group related skills. Each topic includes expository information followed by generic procedures. At the end of the chapters, you'll find a series of practice tasks you can complete on your own by using the skills taught in the chapters. You can use the practice files available from this book's website to work through the practice tasks.

Features and conventions

This book has been designed to lead you step by step through all the tasks you're most likely to want to perform in Microsoft Power BI. If you start at the beginning and work your way through all the procedures, you'll have the information you need to use the program effectively.

However, the topics are self-contained, so you can reference them independently. If you have worked with a previous version of Power BI, or if you complete all the exercises and later need help remembering how to perform a procedure, the following features of this book will help you locate specific information.

- **Detailed table of contents** Search the listing of the topics, sections, and sidebars within each chapter.
- **Chapter thumb tabs and running heads** Identify the pages of each chapter by the colored thumb tabs on the book's open fore edge. Find a specific chapter by number or title by looking at the running heads at the top of even-numbered (verso) pages.
- **Topic-specific running heads** Within a chapter, quickly locate the topic you want by looking at the running heads at the top of odd-numbered (recto) pages.
- **Practice tasks page tabs** Easily locate the practice tasks sections at the end of each chapter by looking for the full-page colored stripe on the book's fore edge.
- **Detailed index** Look up specific tasks and features in the index, which has been carefully crafted with the reader in mind.

You can save time when reading this book by understanding how the *Step by Step* series provides procedural instructions and auxiliary information and identifies on-screen and physical elements that you interact with. The following table lists content formatting conventions used in this book.

Convention Meaning

TIP	This reader aid provides a helpful hint or shortcut to simplify a task.
IMPORTANT	This reader aid alerts you to a common problem or provides information that is necessary to successfully complete a procedure.
SEE ALSO	This reader aid directs you to more information about a topic in this book or elsewhere.
1. Numbered steps 2. 3.	Numbered steps guide you through generic procedures in each topic and hands-on practice tasks at the end of each chapter.
■ Bulleted lists	Bulleted lists indicate single-step procedures and sets of multiple alternative concepts or procedures.
Interface objects	In procedures and practice tasks, bold black text indicates on-screen elements that you should select (click or tap).
Ctrl+P	A plus sign between two keys indicates that you must press those keys at the same time. For example, “press Ctrl+P” directs you to hold down the Ctrl key while you press the P key.
<i>Emphasis and URLs</i>	In expository text, italic formatting identifies web addresses and words or phrases we want to emphasize.

Download the practice files

Although you can complete the practice tasks in this book by using your own instance of Power BI, for your convenience we have provided practice files for some of the tasks. You can download these practice files to your computer from <https://aka.ms/PowerBIDash/downloads>. Follow the instructions on the webpage to install the files on your computer in the default practice file folder structure.

Chapter	Folder	File
1: Introduction to Power BI	Ch01	PBEI_Dashboard.pbix
2: Power BI architecture and administration	None	None
3: Visuals in Power BI	None	None
4: Power BI report development	Ch04	IT Spend Analysis Sample.xlsx IT Spend Analysis Report 01.pbix IT Spend Analysis Report 02 - Formatted.pbix
5: Use DAX in Power BI	None	None
6: Develop Power BI reports from Excel	Ch06	Retail Analysis Sample.xlsx Retail Analysis Sample Report.pbix
7: Develop Power BI reports from SharePoint Online	Ch07	Expense Budgets.stp Expense.stp SharePoint Online.pbix SharePoint Folder.pbix
8: Develop Power BI reports from SharePoint On-Premises	Ch08	Expense Budget.stp Expense.stp Expense - SharePoint On Premises.pbix
9: Develop Power BI reports from SQL	Ch09	SQLSERVER - Sample Report for HR.pbix Sample Report with RLS.pbix
10: Develop Power BI reports from SSAS cube	Ch10	SSAS SampleCube.pbix

11: Develop Power BI reports from Azure SQL	Ch11	SampleReportAzureSQL.pbix
12: Develop Power BI reports from Oracle	Ch12	SampleReport - Oracle.pbix
13: Develop Power BI reports from Dynamics 365	Ch13	Sample-Dynamics 365 Sales Report.pbix

Ebook edition

If you're reading the ebook edition of this book, you can do the following:

- Search the full text
- Print
- Copy and paste

You can purchase and download the ebook edition from the Microsoft Press Store at: <https://www.microsoftpressstore.com/store/microsoft-power-bi-dashboards-step-by-step-9781509308033>.

Get support and give feedback

This section provides information about getting help with this book and contacting us to provide feedback or report errors.

Errata and support

We've made every effort to ensure the accuracy of this book and its companion content. If you discover an error, please submit it to us at <https://aka.ms/PowerBIDash/errata>.

If you need to contact the Microsoft Press Support team, please send an email message to microsoftpresscs@pearson.com.

This page intentionally left blank

Introduction to Power BI

1

In today's competitive marketplace, data is very important for any organization, and it is more important than ever to "know your numbers." If data is represented in an effective manner, it will help organizations make decisions in a timely manner.

Power BI is a powerful business intelligence (BI) tool that enables users to drill into the key metrics of an organization. Businesses use this tool, which generates BI visualizations from specified datasets, to gain key business insights.

When users start working with Power BI, a lot of questions arise about the various licensing options available and the features associated with each version. Selecting the optimal license is key to ensuring success when using Power BI.

This chapter reviews the basics of Power BI and the capabilities supported by the Power BI interface. This includes what Power BI offers, the various Power BI services, the licensing models available, and the process of setting up the environment.

In this chapter:

- Introduction to Power BI
- Power BI services
- Power BI platforms and licensing models
- Set up Power BI
- Prepare a dashboard in Power BI
- Best practices for Power BI security

Practice files

As you work through this chapter, you'll use one practice file, which you can access from the MSPBIDashboards\ch01 folder on the book's companion website at <https://aka.ms/PowerBIDash/downloads>. The file is as follows:

- **PBEI_Dashboard.pbix** This file contains a sample Power BI report used to prepare a dashboard in Power BI service. You'll use this file in the section "Prepare a dashboard in Power BI service" later in this chapter.

An introduction to Power BI

Using business intelligence provides a wider scope for gathering and analyzing data. In this section you will explore what BI is and see a real-world scenario that outlines its possibilities in a general sense. Then you'll get a general overview of Power BI and the different features it offers to users according to their needs.

What is business intelligence?

BI technology enables us to drive meaningful insights from data that can be viewed at various levels of the organization, and to use these insights to drive optimal results. BI consists of theories and methodologies that users can study and apply to make meaningful decisions.

Technologies associated with BI, including Power BI, enable users to easily transform raw unstructured data into more meaningful charts and graphs, making BI a popular tool for transforming business. For example, consider a large organization in which historical data is stored and represented in a form that is difficult for decision-makers to understand. With BI, this data can be converted into a more meaningful form, providing answers to critical questions across the organization.

Real-world scenario

Suppose you work for a popular watch-manufacturing company, and you want to view its historical growth data across the last 30 years. Without BI, the following tasks associated with this effort would have proved difficult:

- Reviewing performance data for each year
- Finding and reviewing the available data to match the customer trend analysis
- Creating interactive visualizations to convey customer-centric goals

With the rise of tools like Power BI, the organization could present the available data into a powerful dashboard with graphs, visualizations, and sales charts, making it easy for members of executive teams to study and drill down on the data. This makes Power BI extremely useful in this critical era of analytics.

What is Power BI?

Power BI is a BI tool developed by Microsoft to present data in a more meaningful way, such as a dashboard, graph, visualization, or report. These are then used to analyze the data to better understand the needs of the organization.

Power BI includes the following capabilities:

- The capability to wrangle data from large databases—even the ones that can read 1 million rows per hour per user
- Dashboards, related visualizations, and reports to present data in a meaningful way for analysis by the end user
- The capability to readily connect faults and errors to provide a corrected methodology and approach for the end user

There are many real-world examples explaining the use of Power BI to spur business growth and transformation. In particular, dashboards and reports generated in Power BI provide actionable analysis to improve the decision-making process and reduce costs.

Who uses Power BI?

Different kinds of people use Power BI in different ways. A general representation is provided in Figure 1-1. For example, one user who studies BI reports might prefer a more enhanced representation of data—for example, with color contrasts and features that affect its look and feel. Another user—for example, a data analyst who studies analytics more often as part of their day-to-day activities—might prefer things the other way around, focusing on the underlying numbers rather than how it is presented in dashboards and related reports.

1	Technical BI IT to End User
2	Analyst to Everyone Analyst to End User
3	End User BI Everyone

FIGURE 1-1 *The three main users of BI.*

Using Power BI effectively to streamline organizational growth

One way for an organization to gain insights for better understanding and collective decision-making is to collect relevant data and streamline it into reports. Consider an engineering firm with expertise in manufacturing precise metal products. A live data update for higher-level staff will provide insight into several aspects of the manufacturing process. The detection of defects in the metal, the percentage of processes that produce these defects, the number of corrections applied to rectify the defects, and so on are just some of the parameters considered for analysis. This dataset can be fed live into the Power BI server to generate meaningful dashboards and reports based upon regular refreshes.

Power BI features

Power BI offers several features. This section reviews these services and discusses how they are used.

Power BI dashboards

Power BI dashboards provide powerful visualizations of datasets in the form of graphs and charts. These can be analyzed in real-time from a live data feed, making them more interactive for users, and enabling them to surface relevant insights quickly.

Power BI dashboards include links, or tiles, to related reports, which helps users navigate from one report to another. These tiles can be easily customized according to user preference. In addition to adding images to these tiles, users can modify how they are laid out on the dashboard.

A word on datasets

A dataset gathers data from various data sources. Generally, there are two types of datasets in Power BI:

- **Dynamic** These datasets refresh automatically according to a set schedule.
- **Static** These datasets do not refresh automatically.

Natural query processing

Power BI users can use a special Question and Answer box in the dashboard to enter a query in plain English. Power BI will then filter, sort, and group data to return a list of visualizations that match the query. For example, suppose a user wants to analyze sales data for their organization. In this case the user could enter queries like, "How many leads have been converted to sales?" and "How many leads have gone cold?" Power BI can quickly provide useful visualizations to answer the user's questions.

User-friendly reports with data visualization

You can use Power BI to generate reports that contain detailed data in eye-catching visual form, along with easy-to-understand analysis of that data. Visualizations in Power BI reports make it easier to comprehend the analysis and arrive at a specific conclusion or decision. Users generate these reports by pinning different parameters to a Power BI dashboard and applying different filters to those parameters to drill down for deeper insights.

A Power BI report might include visuals such as charts with geographical information, bubble charts (which focus on relationships), comparison charts (including bar charts, line charts, waterfall charts, and so on), and many others. You can also consider adding comparison charts and other visuals—for example, to generate tree maps and pie charts.



IMPORTANT To help users create custom Power BI visuals, Microsoft has publicly released the code for these visuals. For more information, see <https://github.com/Microsoft/PowerBI-visuals>.

Sharing with others

All data, reports, and visualizations created by a specific user remain private unless that user decides to share it. Sharing within Power BI is easy. For example, to share a Power BI dashboard with other Power BI users, simply select the sharing option under the Actions category on the dashboard page. Or, to share a report, choose the Share with Others option. Then enter the email address(es) of the desired recipient(s) and specify what permissions they should be granted.

Users can share reports with annotated text and highlights to help end users in the decision-making process. Also, any changes made to a shared report are simultaneously reflected on the dashboard of each person granted access to the report. Users with read-only access can apply different filters for different scenarios to a report, but they cannot alter the report or apply different formatting to it.



IMPORTANT You can only share reports with other users who have purchased or been assigned a Power BI Pro license in the same organization. In other words, the free version of Power BI does not allow sharing. (You'll learn more about the various Power BI licenses later in this chapter.)

Power BI Report Server

Power BI Report Server is an on-premises enterprise reporting and self-service BI feature that provides accurate reporting across organization levels. Users can directly publish Power BI reports on Report Server, thereby enabling other members of the organization to view, manage, edit, or update the reports.



IMPORTANT When you publish a report, you place it on a personal or group workspace where others inside your organization with appropriate permissions can access it. When you share a report, you share it with specific people inside or outside your organization.

Power BI Report Server offers editing tools that enable users to create reports with a modern look and feel. Report Server also generates cloud-ready solutions compatible with SQL Server Reporting Services (SSRS).



IMPORTANT On-premises reporting through Power BI Report Server is available only with the Power BI Premium license. You'll learn more about this license shortly.

Power BI Embedded

Power BI Embedded is a useful feature that enables you to embed Power BI reports into materials generated by third-party apps or in emails to share them with non-Power BI users. In addition, data in the form of graphs or other visuals can be rendered in any browser, irrespective of the dataset chosen. End users receive up-to-date analysis of data developed and/or stored within the Microsoft Azure framework. Microsoft suggests that in the near future, Power BI Embedded will become a preferred method for users using a Premium license. (License options are discussed later in this chapter.)

According to Microsoft, Power BI Embedded has some limitations:

- It cannot be used with any on-premises data gateway. This becomes difficult for developers working on specific requirements or customizations.
- Users who have access to Power BI cannot work on the same namespace, as Power BI Embedded runs on a completely different premium capacity model.

Mobile support

The Power BI mobile app enables you to access Power BI dashboards and reports using a mobile device such as an Android or iOS device. In some scenarios analysts can use the mobile app to address minute changes while tracking data in real-time and notify users of a possible solution.

Note that it is sometimes difficult to view BI reports on smaller screens; fortunately, Power BI allows for optimal display at correct resolutions. You can also zoom in and out and drill down to see relevant information feeds within the charts and graphs.

Power BI platforms

Power BI offers users the ability to choose from various platforms depending on their needs:

- Power BI service
- Power BI Desktop
- Power BI Premium

Power BI service

Power BI service is a free, cloud-based platform that works on Microsoft Azure. This version of Power BI is not available on a private cloud or as an internal service.

Power BI service includes the following capabilities and features:

- The capability to connect with hundreds of data sources
- An online dashboard with analytics information including graphs and reports
- Customizable dashboards
- Customizable reports
- The capability to edit reports online
- The capability to easily share reports
- Navigation controls that allow easy access to datasets
- Workspaces to collaborate with others on dashboards, reports, and datasets to create apps

- Home, Help, and Feedback buttons
- An Office 365 app launcher
- A Question and Answer box that enables users to search for data using plain English

Power BI Desktop

Power BI Desktop is like Power BI service except that it lives on the user's desktop machine and not in the cloud. This free tool integrates seamlessly with queries, data modelling, and visualizations for effective and accurate reporting of data, and offers a wide range of features to enable users to create appealing visualizations and detailed reports that can be published directly to other apps, to the web, or for mobile view. Note, however, that Desktop users cannot share reports.

Advantages of using Power BI Desktop include the following:

- Power BI Desktop supports various data sources, which is helpful in adapting to complex business requirements.
- Power BI Desktop includes an auto-detect relationship option for loaded datasets to help users gain understanding and insights.
- A user-friendly interface facilitates the creation and editing of custom visuals.
- Users can save reports for easy access. Reports that are saved have a PBIX extension.
- Users can create and publish reports from a single Power BI Desktop dashboard.

Power BI Premium

Power BI Premium provides greater capabilities and performance by increasing capacity-based offerings to end users. These act as dedicated resources to run the Power BI service for any organization. This dedicated capacity provides better support and stability.

Power BI Premium enables widespread distribution of data across the enterprise—without requiring additional licenses for users within the organization. A Premium workspace is a dedicated workspace allocated within the organization. It works on dedicated hardware handled by Microsoft for bulk data interaction per hour. The workload and number of users change based on specific factors and dictate how many users can access and edit data for reports. In this way, the capacity-based

resources work in coordination to handle reports, graphs, and visualizations across the entire enterprise level.

Embedded analytics can be accessed from the Azure API, which offers the flexibility to obtain live data. The Power BI Report Server helps in maintaining BI assets on-premises. The Power BI Premium Report Server supports compatibility with SQL Server Reporting Services (SSRS) to provide interactive visualizations and paginated reports that can be analyzed on-premises. Report Server, which is a part of Power BI Premium, continuously deploys data from the dataset for better graphical analysis of reports, which can later be moved to the cloud.



IMPORTANT Premium removes the down limits for refreshes and dataset sizes, which thereby increases the stability and performance for isolation with a dedicated hardware.

Power BI licensing models

Microsoft offers three types of licenses for Power BI:

- Power BI Free
- Power BI Pro
- Power BI Premium

The features of each license vary. The one you choose depends on your business requirements, including how much storage you need and the total number of users. The following sections consider the various Power BI licensing models.

Power BI Free

Power BI Free enables users to create any number dashboards and reports. However, they cannot share these with others.

Power BI Pro

Power BI Pro offers features that are not available in the free version platform. Most notably, Power BI Pro enables users to share dashboards, enabling multiple users to access them at the same time. Users can also collaborate to create any number of graphs and reports.

Other advantages of using Power BI Pro include the following:

- Users can connect to more than 70 data sources.
- Users can analyze data in the Power BI dashboard or in an Excel file.
- Users can synchronize content across Office 365 teams.
- Users can customize views according to their preferences.
- Users can develop personal dashboards, which they can save and even pin to their Power BI home page.
- Users can access dashboards shared by other members of the Office 365 community (assuming they have appropriate permissions) for collaboration purposes.
- Users can share workspaces.
- Users can subscribe to email alerts to notify them of changes to shared content.
- Users can embed APIs into Power BI visuals and reports.
- Users can publish reports and visuals to a web platform.

Here are a few examples of scenarios that require a Power BI Pro license:

- To publish content in Power BI Premium
- To share and receive content from other users
- To collaborate with other users on a dashboard or report
- To publish reports on Power BI Report Server

Power BI Premium

Power BI Premium is similar to Power BI Pro but can be scaled for larger organizations to accommodate more users or to provide users with more storage or a higher stream rate.

Power BI Premium offers the following advantages:

- Power BI Premium unlocks larger data limits and higher capacity offerings to accommodate large-scale deployments.
- Power BI Premium allows you to embed reports in other applications. (Note that for REST APIs to work with embedded graphs and reports, they will use a service account, which is available in Power BI Premium.)

- The Power BI Premium Report Server allows for the publication of reports on-premises.
- Dedicated servers allow for enhanced performance, with data refreshing 48 times per day.

Comparing the Free, Pro, and Premium licenses

Table 1-1 outlines key differences between the Power BI Free, Pro, and Premium licenses.

Table 1-1 *Power BI Free and Pro versus Power BI Premium*

Power BI Free and Power BI Pro	Power BI Premium
Storage is limited to 1 GB per user (Free) or 10 GB per user (Pro).	Ability to publish at premium capacity.
Data is refreshed 8 times per day.	Data is refreshed 48 times per day.
Data can be handled on-premises and in the cloud.	Data is secured with organizational content packs.
Data streaming occurs at a rate of 1 million rows per hour.	The data-streaming rate depends on the capacity defined for the report or workspace.
Necessary service requests are supported.	Row Level Security (RLS) is available for all users.
Data can be sourced from live datasets.	Data is available in the form of data catalogs.

Set up Power BI

Power BI is an effective reporting and visualization tool that enables you to generate dynamic dashboards and reports. There are numerous options available to you when you sign up for Power BI. You can sign up for a free version or purchase a subscription for Power BI Pro. (You can also sign up for a free 60-day trial of Power BI Pro.) The free account provides all the functionality of Power BI Pro except for sharing and collaboration features.



IMPORTANT There are no options within the free trial to upgrade to a Power BI Premium license. Users opting for this setup will need to purchase it directly from the official Microsoft Power BI link.



IMPORTANT You must use a work email address to sign up for Power BI. Personal email addresses aren't allowed.

1

Set up Power BI service

To set up Power BI service, follow these steps:

1. Type <https://powerbi.microsoft.com> in your web browser's address bar to open Microsoft's Power BI page.
2. Click the **Sign Up Free** link in the top-right corner of the screen.
3. Click the **Try Free** button under the Power BI heading.
4. When prompted, enter your work email address. Microsoft will send a verification code to that email address.
5. A Create Your Account dialog box opens. Enter your first name and last name, set and confirm a password, and type the verification code sent to the email address you provided in step 4. (See Figure 1-2.) Then click the **Start** button.

Create your account

First name Last name

Create password

Confirm password

Verification code [resend signup code](#)

Microsoft will send you promotions and offers about Microsoft products and services for businesses. You can unsubscribe anytime.

By choosing Start, you agree to our [terms and conditions](#) and understand that your name and email address will be visible to other people in your institution. [Microsoft Privacy Policy](#)

Start ➔

FIGURE 1-2 Signing up for Power BI service.

- 6. A message appears instructing you to invite more members of your organization to create Power BI accounts. If desired, enter the email addresses of anyone you'd like to invite and click **Send Invitations**. Alternatively, click **Skip** for now.

Power BI service opens, displaying the dashboard with a welcome message. (See Figure 1-3.)

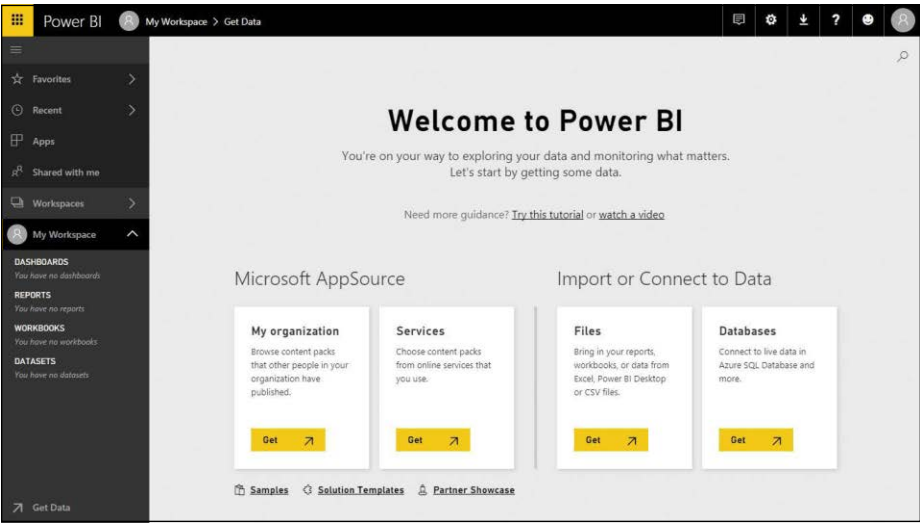


FIGURE 1-3 The Power BI service welcome dashboard.

Set up Power BI Desktop

To run Power BI Desktop, your system must meet certain minimum requirements. These appear in Table 1-2.

Table 1-2 Minimum requirements to install Power BI Desktop

Operating System	Windows 7/Windows Server 2008 R2 or later
.NET Framework	.NET 4.5
Browser	Internet Explorer 9 or later
RAM	At least 1 GB; 1.5 GB or more recommended
Display	At least 1440x900 or 1600x900 (16:9) recommended
CPU	1 GHz or faster; x86- or x64-bit processor recommended

To download and install Power BI Desktop, follow these steps:

1. Type <https://app.powerbi.com> in your web browser's address bar.
2. Click the **Sign In** button and sign in using your work email credentials.
3. Click the **Download** button in the top-right corner of the screen and select **Power BI Desktop** from the menu that appears. (See Figure 1-4.)

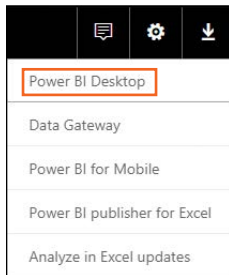


FIGURE 1-4 Downloading Power BI Desktop.



TIP Another way to start the download process is to direct your web browser to <https://powerbi.microsoft.com/en-us/downloads/> and click the **Download** button under Microsoft Power BI Desktop.

4. After the installation file is downloaded, right-click and choose **Run as Administrator** to begin the setup process.
5. The Microsoft Power BI Desktop wizard starts. Click the **Next** button. (See numeral 1 of Figure 1-5.)
6. Select the **Terms and Conditions** checkbox and click **Next**. (See numeral 2 of Figure 1-5.)
7. Leave the file path set to the default. Then click **Next**. (See numeral 3 of Figure 1-5.)
8. If you like, select the **Create a Desktop Shortcut** checkbox. Then click **Install**. (See numeral 4 of Figure 1-5.)
9. Optionally, after the installation is complete, select the **Launch Microsoft Power BI Desktop** checkbox. Then click **Finish**. (See numeral 5 of Figure 1-5.)
10. The Power BI Desktop Sign In page appears (See numeral 6 of Figure 1-5.) Click the **Sign In** button and sign in to open Power BI Desktop.

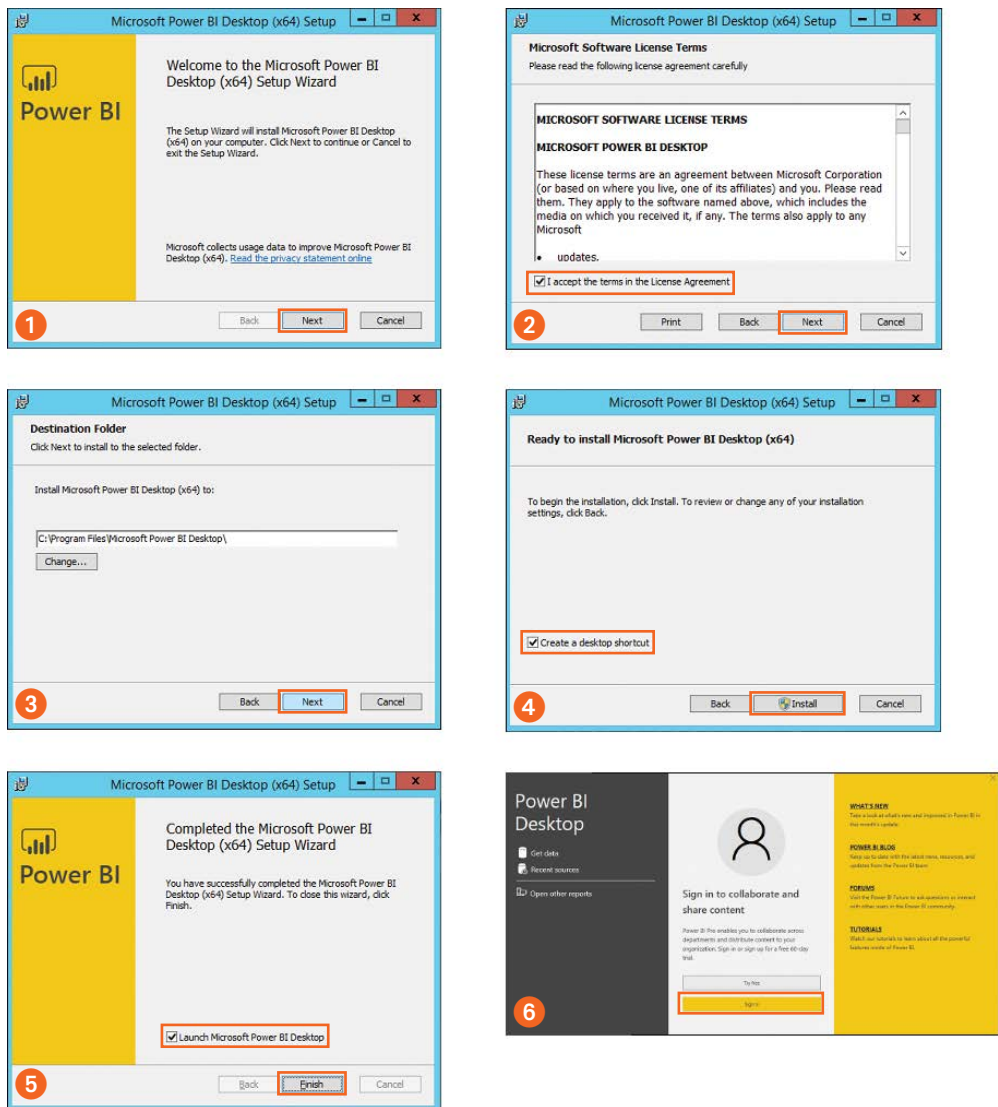


FIGURE 1-5 Installing Power BI Desktop.

Set up Power BI on your mobile device

In addition to using Power BI on a desktop machine, you can also use it on your mobile phone, thanks to the Power BI mobile app. This app is available for the following mobile platforms:

- Windows (Windows Store)
- Android (Google Play Store)
- iOS (Apple App Store)

To install Power BI on an Android device, follow these steps. (The steps to install the app on a Windows or iOS device are similar.)

1. Depending on what type of mobile device you have, open the Windows store, the Google Play store, or the Apple App store on your device.
2. Search for the Microsoft Power BI app.
3. Tap the Install button to install the Power BI app on your phone. (See numeral 1 of Figure 1-6.)
4. Open the app. As shown in numeral 2 of Figure 1-6, you'll see two options:
 - Power BI
 - Report Server (part of Power BI Premium)
5. Tap **Power BI**.
6. Enter your work email address and your password and tap **Sign In**. (See numeral 3 of Figure 1-6.) A dashboard will open.

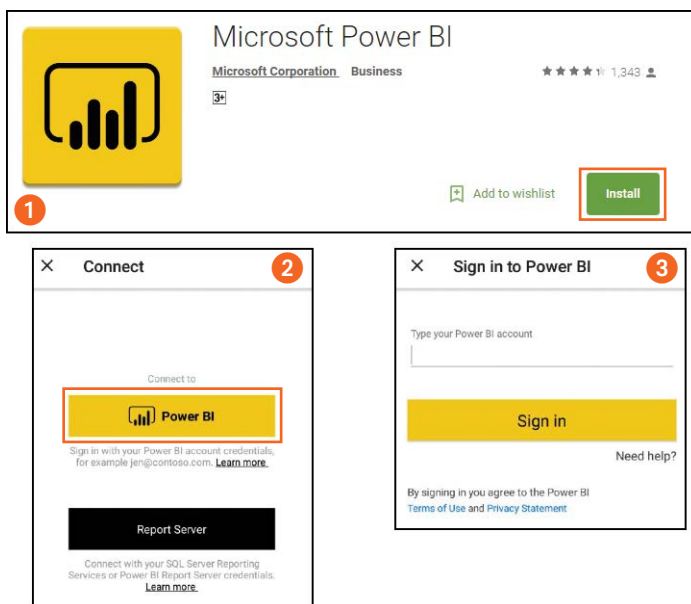


FIGURE 1-6 Setting up Power BI on a mobile application.

Prepare a dashboard in Power BI service

Power BI users can prepare custom dashboards. This section shows you how to prepare a simple custom dashboard based on a report prepared and published in Power BI service that uses various out-of-the-box visuals available in Power BI service. (You'll explore a detailed step-by-step approach to preparing reports using different data sources later in this book.)



IMPORTANT Before you can prepare a dashboard based on a Power BI report, you must publish the report. To find out how to publish a report, see Chapter 4. Once you've published a report, you can follow the steps in this section.

Follow these steps to build a dashboard in Power BI service:

1. Type `https://app.powerbi.com` in your web browser's address bar.
2. Click the **Sign In** button and sign in using your work email address. As shown in Figure 1-7, Power BI service opens with an empty workspace displayed. (You'll learn more about workspaces in Chapter 2.)

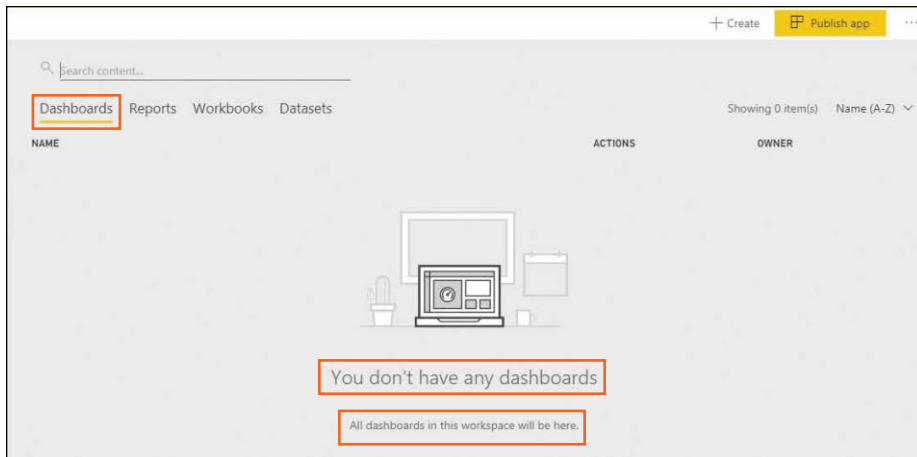


FIGURE 1-7 An empty Power BI workspace.

- To view reports that have been prepared and published in Power BI service, click the **Reports** option in the workspace and select the report from which you want to build a dashboard.
- Open the report you want to use. (In this example, I used the sample file I mentioned at the beginning of the chapter, PBEI_Dashboard.pbix.)

As shown in Figure 1-8, this report contains two visuals: PTQ Trend and Sales by Device Type.

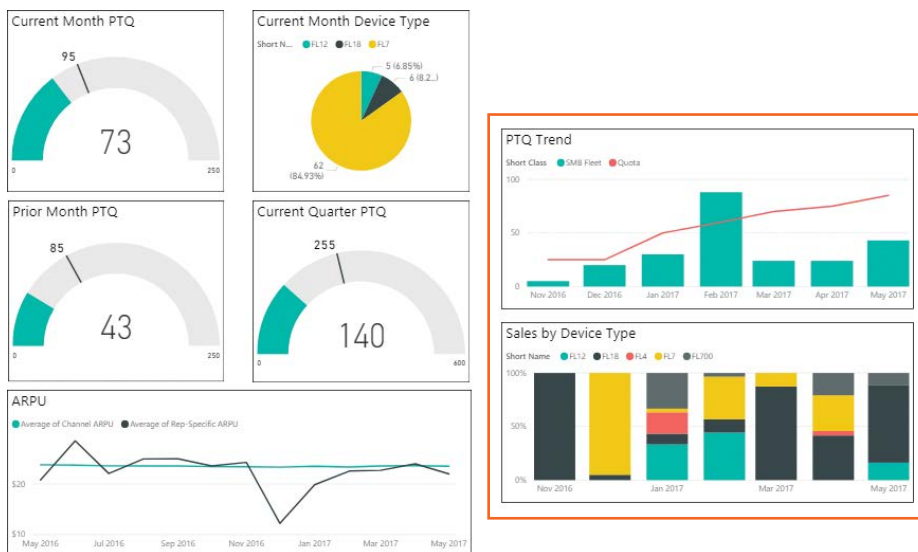


FIGURE 1-8 Sample Power BI report.

- Click the **Pin Visual** button in the upper-right corner of the PTQ Trend visual to pin it to the dashboard. (See Figure 1-9.)

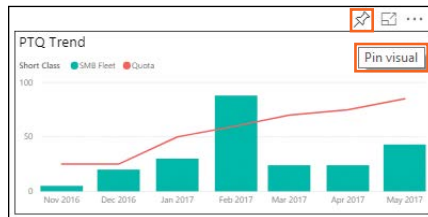


FIGURE 1-9 A sample Power BI visual with an option to pin the visual to a dashboard.

A Pin to Dashboard dialog box opens. (See Figure 1-10.)

- Select the **New Dashboard** option button, type a name for the new dashboard, and click the **Pin** button. Power BI notifies you that the visualization was successfully pinned.

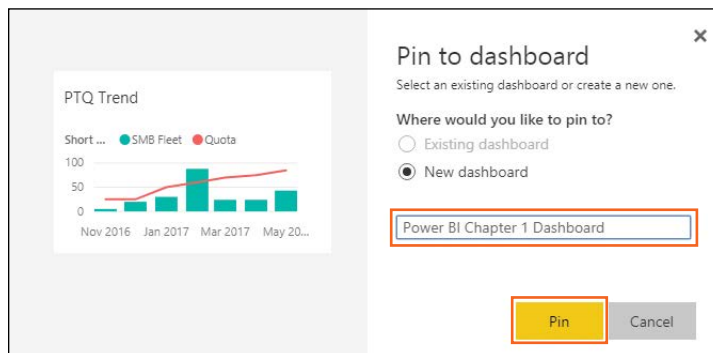


FIGURE 1-10 The Pin to Dashboard dialog box.

- Repeat these steps to add more visualizations to the dashboard, such as the **Sales by Device Type** visualization. This time, however, choose **Existing Dashboard** in the Pin to Dashboard dialog box and select the desired dashboard. (See Figure 1-11.) Figure 1-12 shows a dashboard with several visuals added.

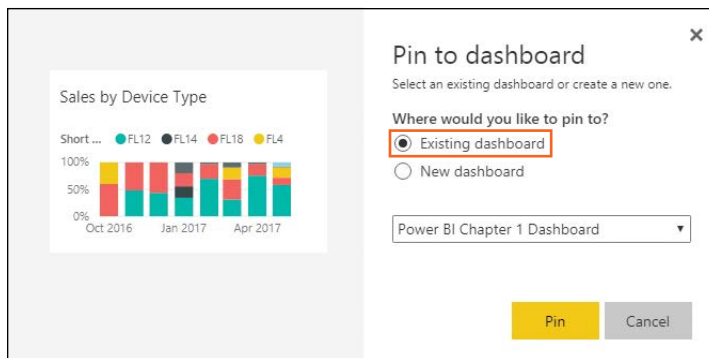


FIGURE 1-11 Pinning a visualization to an existing dashboard.



FIGURE 1-12 The new Power BI dynamic dashboard.

General note for Power BI security

Microsoft Azure is a cloud-based software as a service (SaaS) platform that empowers users to easily build and manage cloud-based integrations online. Because Power BI is built on the Azure framework, it's fairly easy to secure it from unauthorized users.

In general, Power BI security services depend on two clusters: the web front end (WFE) and the back-end cluster. Using the WFE, the user enters their login credentials. Power BI then uses Azure Active Directory (AAD) to authenticate the user. After the user is authenticated, the back-end cluster takes over. It handles data reporting, manages dashboard data, takes care of data connections, and manages live datasets. Basically, it handles everything from the time the user enters a request until the Power BI service responds to that request.

Skills review

In this chapter, you learned about:

- Business intelligence (BI)
- How Power BI effectively streamlines organizational growth
- Features of Power BI
- Power BI platforms and licensing models
- Procedures for setting up Power BI service, Power BI Desktop, and Power BI for mobile devices
- The procedure for building a dashboard in Power BI

Practice tasks

This section provides a simple case study for you to study and solve using information from this chapter. This section also contains a series of practice questions for you to answer.



Case study

Consider a large finance company in which an IT project manager is charged with deciding which version of Power BI to adopt. Here's the scenario:

- The finance organization has millions of rows of data.
- Thousands of new rows are added to the database every hour.
- Employees have various roles, according to their professions.
- Some employees act only as end users, viewing reports containing this data.
- Some employees are analysts and require the ability to share Power BI dashboards.
- Some employees focus on high-precision data to improve their understanding of performance.
- The finance company wants to set up an on-premises environment within the organization.

Based on these points, decide whether the finance company should select the Power BI Pro version or the Premium package.

Practice questions

1. What is the difference between Power BI Pro and Power BI Premium?
2. What services does Power BI support?
3. Describe the natural query processing capabilities in Power BI.
4. What Power BI feature supports an on-premises environment?
5. What is Power BI Report Server?
6. What is Power BI Embedded?

7. What kind of license do you need to share your Power BI dashboard with other team members?
8. Why are visuals used in Power BI Desktop?
9. How do you set up Power BI Desktop?
10. How do you set up a dynamic dashboard in Power BI?

Power BI architecture and administration

2

Power BI is one of the most valuable tools available for organizations that seek to gain key insights from their data. Power BI enhances end user efficiency, improves data analysis, and facilitates the overall decision-making process.

This chapter covers the architecture and the underlying services of Power BI, including the internal and external working structures. It also takes a look at Power BI administration processes, in which an assigned user has specific roles to monitor and control the use of Power BI.

This chapter then explains data modeling functionality in detail—the process of converting raw data into a simpler form that can be read by Power BI servers—as well as the report-development process in Power BI Desktop and Power BI service (including workspace functionality).

In this chapter:

- Power BI architecture
- The Power BI Admin Portal
- Power BI data models and report development

Power BI architecture

To better understand Power BI operations, it's important to have a grasp of its architecture from both a technical and an internal standpoint. That's the focus of this section. This section also provides an overview of the Power BI data-processing model.

Power BI technical architecture

By Power BI technical architecture, we mean the manner in which Power BI ingests data, processes it, and spits it back out for analysis. Figure 2-1 illustrates this architecture.

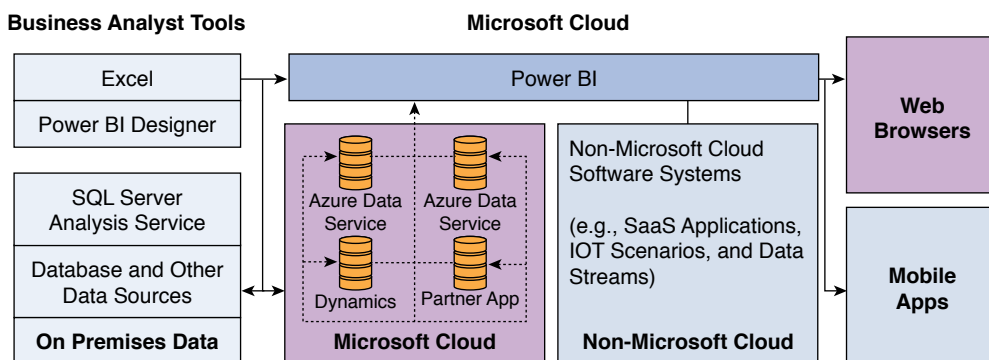


FIGURE 2-1 Power BI technical architecture.

As shown in Figure 2-1, the architecture operates as follows:

1. Power BI servers compile raw data from various sources, including Excel, Power BI Designer, SQL Server, and other database servers, as well as Microsoft Cloud sources such as Azure data services, Microsoft Dynamics, non-Microsoft SaaS apps, and others.
2. Power BI looks for direct relationships among the data fed into the system to develop a sequential data model. (Correlations can also be made manually.)
3. Power BI transforms the data model into visualizations and reports for end users. These users can publish the reports and visualizations online or on various desktop or mobile applications for consumption by other users. Depending on what licensing model of Power BI is used, they may also be able to share them with other users for collaboration purposes.

Power BI internal architecture

The internal architecture of Power BI revolves around the sources of data it ingests. For example, suppose you have an on-premises database that contains SharePoint content, which is fed directly into Power BI Desktop. You also have data sources such as on-premises data files, data on OneDrive for Business, data in Excel files, data in Azure, and more, which enter your system through a personal gateway, after which they are filtered for use by Power BI. The Power BI system can use all the data in available datasets to generate dashboards and reports. Users can pin these reports on their Power BI dashboard and (depending on what licensing model of Power BI they use) share them with others. They can also consume these reports online using Power BI service or via the Power BI Windows Phone, Android, or iPhone app. (See Figure 2-2.)

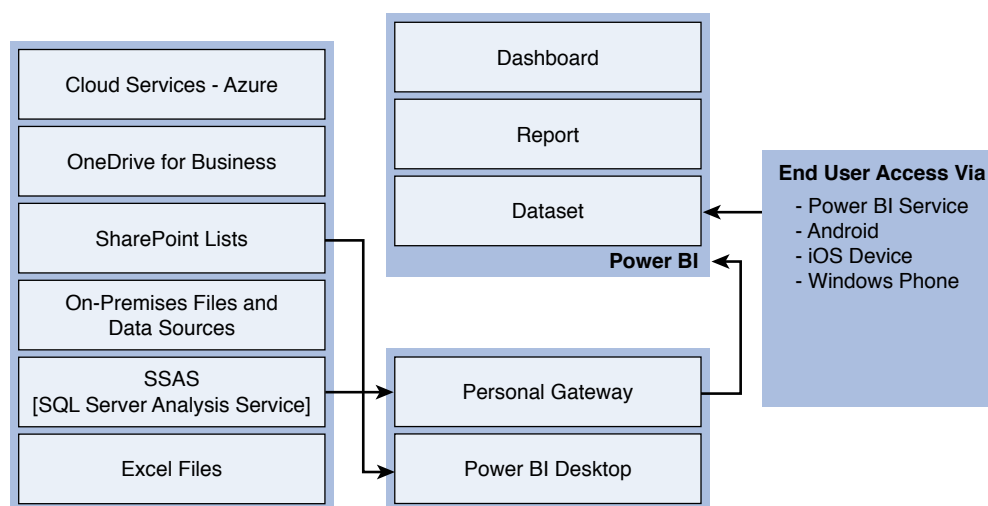


FIGURE 2-2 *Power BI internal architecture.*



IMPORTANT The process by which data sources, in the form of datasets, interact with Power BI is called the conception data flow. The interaction between end users and Power BI is called the visualization data flow. The simultaneous process that occurs between datasets, Power BI, and user-generated reports is called the refresh flow.

Data processing in Power BI

The data ingested by Power BI goes through various processes to ensure its integrity. This is an important part of Power BI analysis. Figure 2-3 outlines these processes.

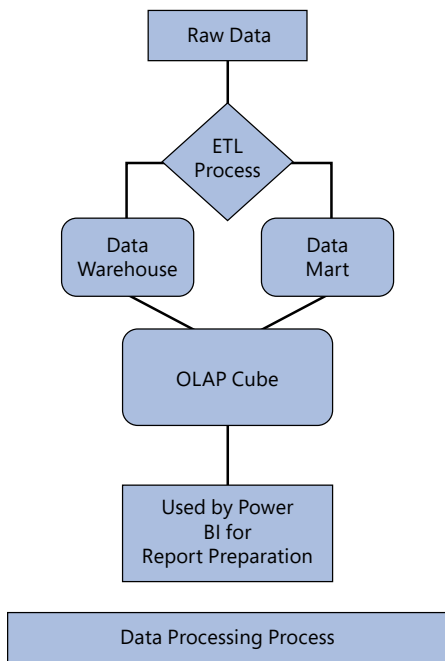


FIGURE 2-3 *Data-processing in Power BI.*

As you can see in Figure 2-3, the processes are as follows:

1. Raw data passes through the extract, transform, load (ETL) process, during which the data is read and assigned a specific data type.
2. The data types rearrange themselves to provide meaningful information in the form of a data warehouse or data mart.
3. The data warehouse and data mart create online analytical processing (OLAP) cubes. An OLAP cube is an array of multidimensional time-specific information that is stored and processed for generating reports and analytical visualizations.
4. The OLAP cubes are fed into Power BI, and their data is converted into maps, histograms, graphs, charts, and other visuals to generate reports.



IMPORTANT Throughout these processes, the integrity and accuracy of the data are maintained—meaning you get near-perfect data in Power BI reports.

These processes—and by extension the Power BI architecture—can be divided into three main phases, or layers. (See Figure 2-4.)

- **Data preparation layer** In this phase, or layer, the system data sources synchronize data. At first, the data sourced from the system exists in different file formats. To read these various sources of information, Power BI first employs a gateway to process the data and then integrates it into a dataset.
- **Data warehouse layer** In this layer, a data warehouse applies filters to the dataset to process it. To maintain accuracy, the data warehouse also uses ETL processes to identify flaws and make corrections as needed. It then forwards the processed data to a Power BI OLAP storage unit.
- **Data presentation layer** After the dataset has been processed by the data warehouse and stored in an OLAP storage unit, it is ready to be converted into reports and visualizations by Power BI. Users can access these visualizations on their dashboard.

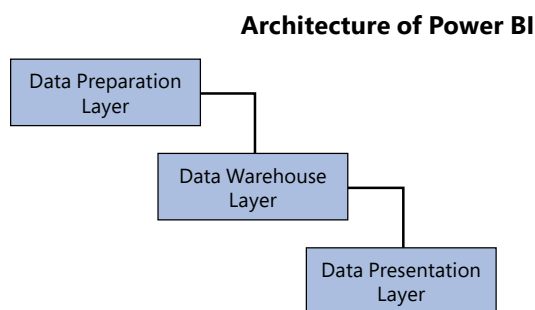


FIGURE 2-4 Three main phases of Power BI architecture.

The Power BI Admin Portal

You can use the Power BI Admin Portal to perform various administrative tasks. To access the Admin Portal, follow these steps. (Note that these steps assume you are signed into your Office 365 account.)

1. Type <https://app.powerbi.com> in your web browser's address bar.
2. Click the **Sign In** button and sign in using your registered work email address.
3. Click the **Settings** icon (the one with the gear) in the top-right corner of Power BI service dashboard and choose **Admin Portal** in the menu that appears. (See screen 1 in Figure 2-5.) The Admin Portal opens. (See screen 2 in Figure 2-5.)

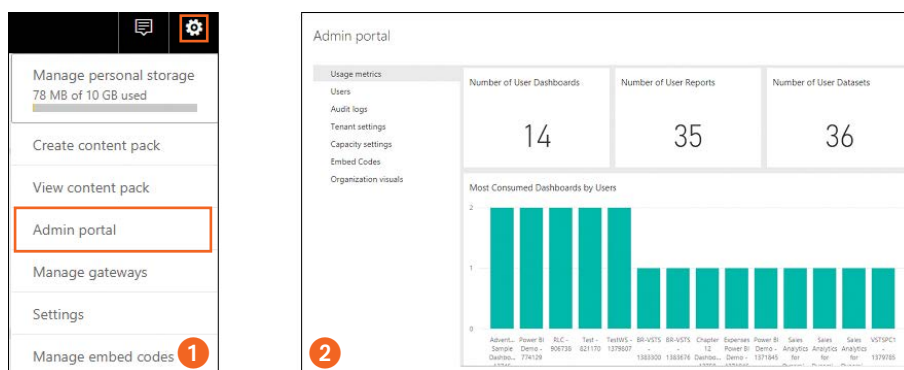


FIGURE 2-5 Accessing the Power BI Admin Portal.



IMPORTANT To understand various modes of Power BI administration, you need to configure and view the Office 365 Admin Portal. This allows admin roles to be assigned to different users.

Assign the admin role to another user

To assign the Power BI administrative role to different users, follow these steps:

1. Type <https://portal.office.com/adminportal/home#/homepage> in your web browser's address bar.
2. The Admin Portal opens. In the pane on the left, click **Users** and select **Active Users**. (See screen 1 in Figure 2-6.)
3. Select the user to whom you want to assign a specific role.
4. Click the **Edit** button. (See screen 2 in Figure 2-6.)
5. Select the **Customized Administrator** option button. Then select the **Power BI Service Administrator** checkbox. (See screen 3 in Figure 2-6.) Then save your changes.

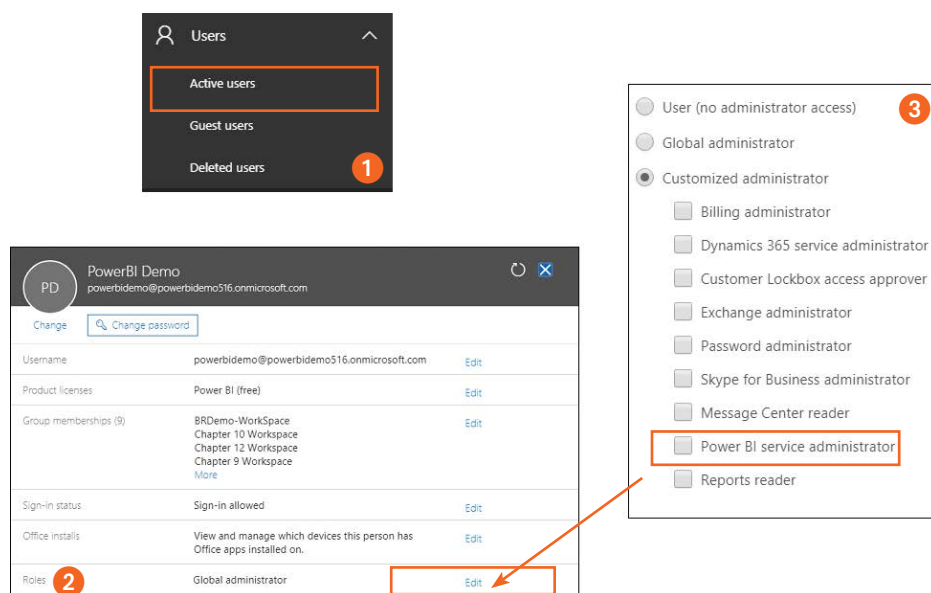


FIGURE 2-6 Assigning the administrator role to a different user.

Explore Admin Portal features

The Power BI Admin Portal consists of several features. These include the following:

- Usage Metrics
- Users
- Audit Logs
- Tenant Settings
- Capacity Settings
- Embed Codes
- Organization Visuals

You access these features from links in the navigation pane on the left side of the Admin Portal. (Refer to Figure 2-5.)

Usage Metrics

The Admin Portal's Usage Metrics page shows a rich dashboard composed of different reports from throughout the organization. The purpose of this page is to provide analysis and insight into a Power BI report.

The Usage Metrics page enables you to see which users and groups are most active within the organization. As shown in Figure 2-7, available usage data for individual users includes the following:

- **Number of User Dashboards** This shows the number of dashboards in the user's workspace. (See tile 1 in Figure 2-7.)
- **Number of User Reports** This shows the number of reports in the user's workspace. (See tile 2 in Figure 2-7.)
- **Number of User Datasets** This shows the number of datasets in the user's workspace. (See tile 3 in Figure 2-7.)
- **Most Consumed Dashboards by Users** This shows which dashboards have been consumed by the most users. (See tile 4 in Figure 2-7.) For example, if someone builds a dashboard, and 15 other people in an organization share it, and 10 more people share its content pack, then the total number of users who have consumed that dashboard is 26 (1+15+10). (A content pack is a collection of dashboards, reports, datasets, and workbooks that are combined together within an organization.)
- **Most Consumed Packages by Users** This shows the most popular content with which the user is associated. These packages can be anything from SaaS content packs, to organizational packs, to files, to a database. (See tile 5 in Figure 2-7.)
- **Top Users with Most Dashboards** This lists the users who have created (or consumed through sharing) the most dashboards. (See tile 6 in Figure 2-7.)
- **Top Users with Most Reports** This lists which users have created (or consumed through sharing) the most reports. (See tile 7 in Figure 2-7.)

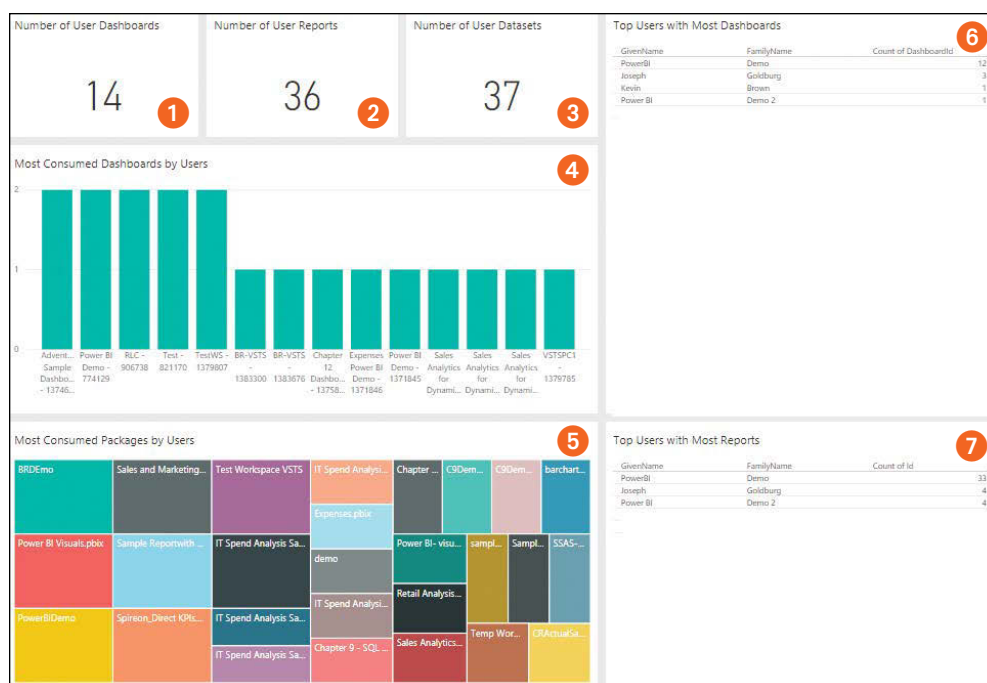


FIGURE 2-7 Usage metrics for individual users.

In addition, you can view the following usage data for groups. (See Figure 2-8.)

- **Number of Group Dashboards** This shows the number of dashboards in the group's workspace. (See tile 1 in Figure 2-8.)
- **Number of Group Reports** This shows the number of reports in the group's workspace. (See tile 2 in Figure 2-8.)
- **Number of Group Datasets** This shows the number of datasets in the group's workspace. (See tile 3 in Figure 2-8.)
- **Most Consumed Dashboards by Groups** This shows which dashboards have been consumed by the most groups. (See tile 4 in Figure 2-8.)
- **Most Consumed Packages by Groups** This shows which packages have been consumed by the most groups. (See tile 5 in Figure 2-8.)

- **Top Groups with Most Dashboards** This lists the groups that have created (or consumed through sharing) the most dashboards. (See tile 6 in Figure 2-8.)
- **Top Groups with Most Reports** This lists the groups that have created and viewed the most reports. (See tile 7 in Figure 2-8.)

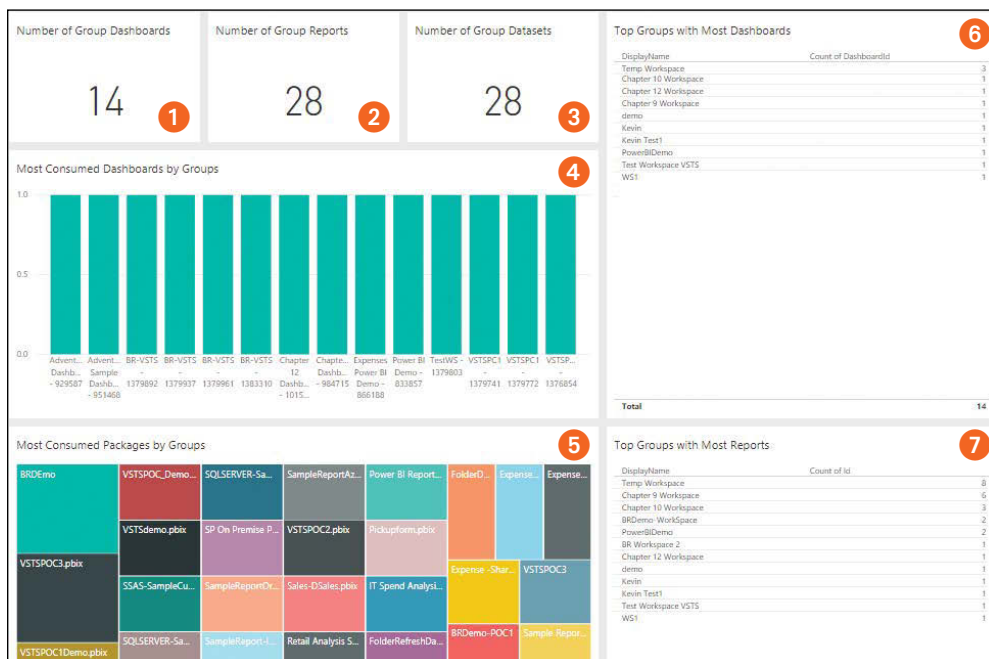


FIGURE 2-8 Usage metrics data for individual groups combined.

Users

The Users page provides access to tools to manage users, administrators, and groups in the Office 365 Admin Center. To access these tools, simply click the **Go to O365 Admin Center** button in the Users page. (See screen 1 in Figure 2-9.) This opens a page in the Office 365 Admin Center that shows all active users with proper administration controls for Power BI. (See screen 2 in Figure 2-9.)

2

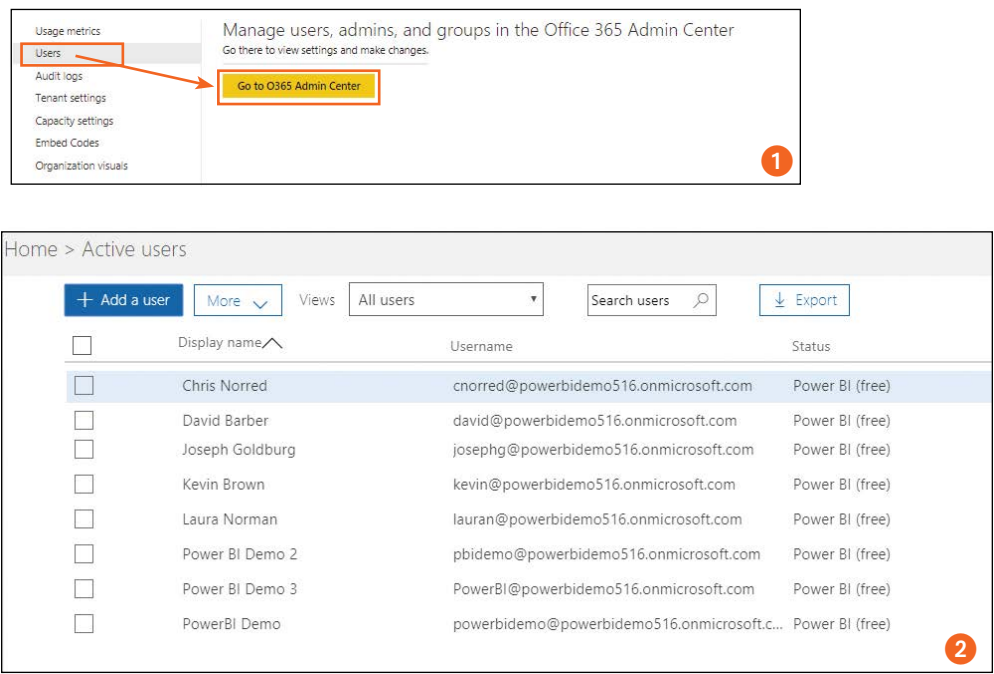


FIGURE 2-9 Accessing tools for managing users.

Audit Logs

Power BI enables you to view audit logs for various reports. To do so, click the **Go to O365 Admin Center** button in the Audit Logs page. (See screen 1 in Figure 2-10.) This opens the Office 365 Security and Compliance Center, where you can search for the audit log you want to see. (See screen 2 in Figure 2-10.)

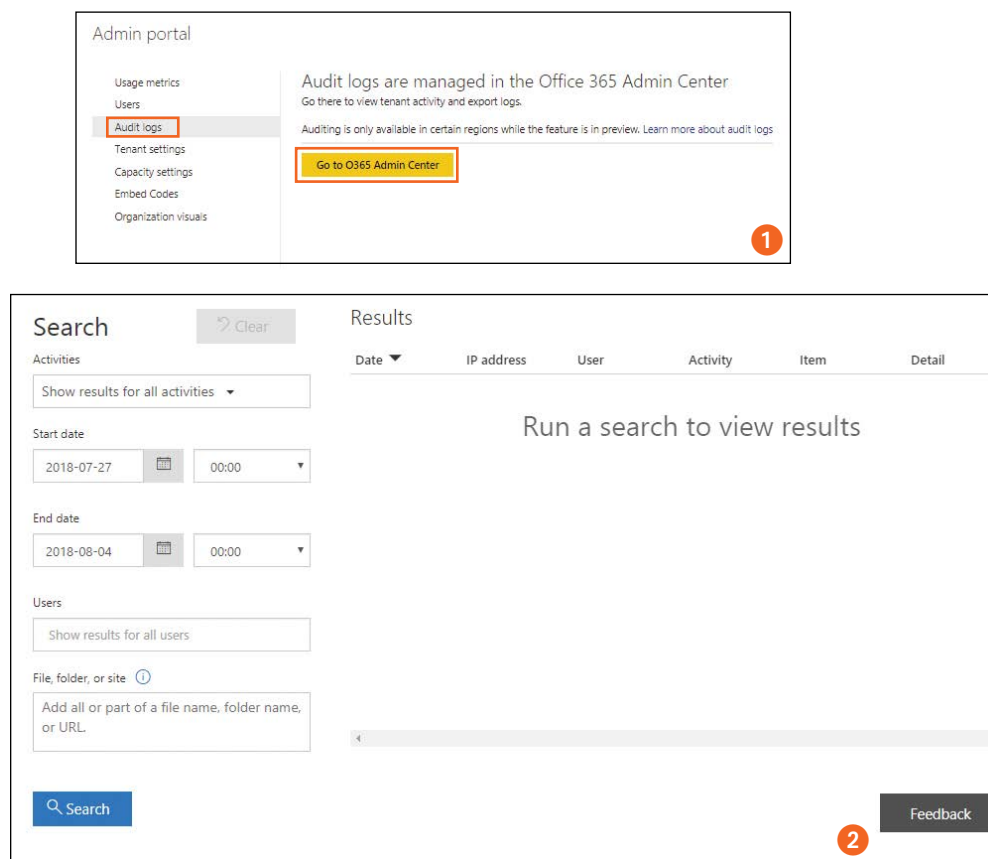


FIGURE 2-10 Viewing audit logs.

Tenant Settings

A tenant is the collection of entities within an organization, such as users, domains, subscriptions, and so on. Applying tenant settings enables you to control who in the organization can access what and facilitates the sharing of sensitive data with specific groups. There is also a feature to switch various tenant-level settings on or off based on user needs.

Power BI supports the following tenant settings:

- Export and sharing settings
- Content pack and app settings
- Integration settings
- R visuals settings
- Audit and usage settings
- Dashboard settings
- Developer settings

Export and sharing settings

The Export and Sharing Settings page includes several groups of settings to control how content is exported and shared. These include the following:

- **Share Content with External Users** Settings in this group control whether members of your organization can share content with external users. Your options are as follows. (See screen 1 in Figure 2-11.)
 - **Disabled/Enabled** Click this toggle to disable or enable this feature. Enabling this feature engages the following settings:
 - **Enable for Entire Organization** Select this option button to enable the feature for all the people in an organization.
 - **Specific Security Groups** Select this option button to enable the feature for specific groups in the organization.
 - **Except Specific Security Groups** Select this option button to prevent the setting from being applied to a specific group.
- **Publish to Web** Settings in this group dictate whether members of your organization can publish Power BI dashboards and reports to the web. This group contains the same settings as the Share Content with External Users group. (See screen 2 in Figure 2-11.)
- **Export Data** Settings in this group indicate whether members of your organization can export data from a tile or visualization. Again, this group contains the same settings as the others. (See screen 3 in Figure 2-11.)

- **Export Reports as PowerPoint Presentations** Settings in this group control whether members of your organization can export Power BI reports as PowerPoint presentations. Again, this group contains the same settings as the others. (See screen 4 in Figure 2-11.)
- **Print Dashboards and Reports** Settings in this group control whether members of your organization can print dashboards and reports. (See screen 5 in Figure 2-11.)

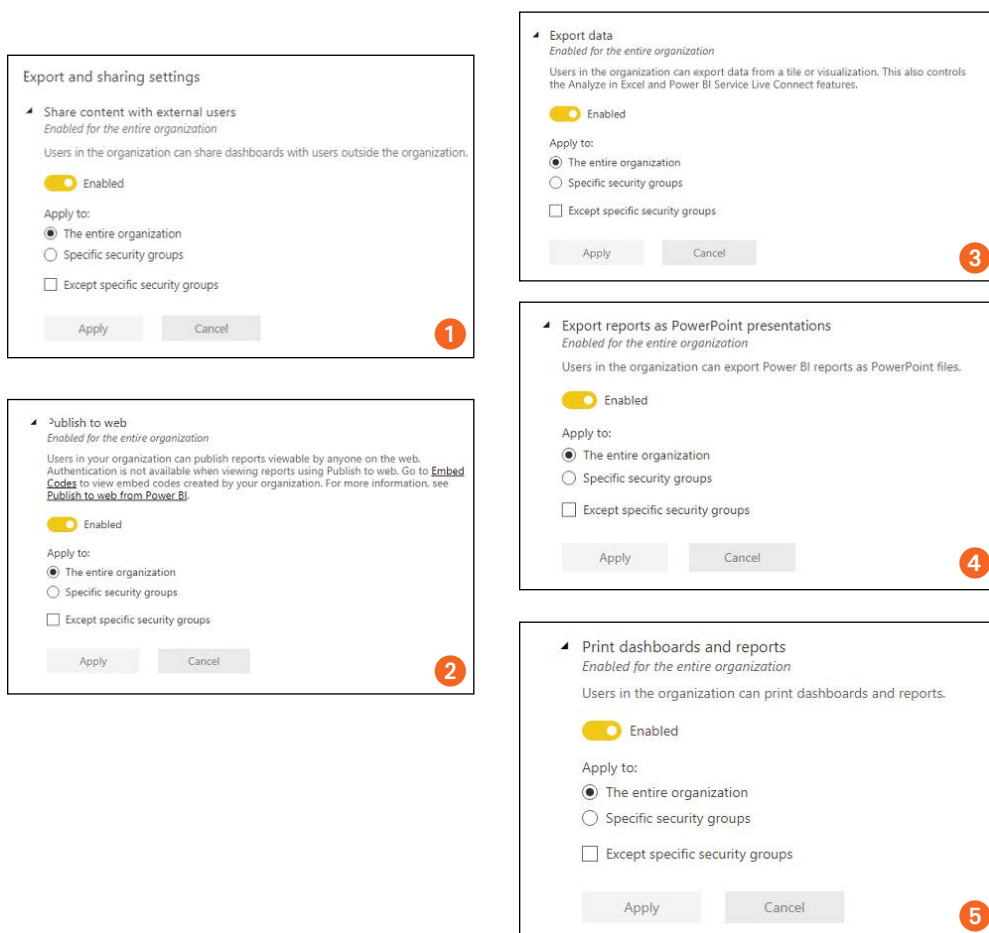


FIGURE 2-11 Tenant-level settings in the Power BI Admin Portal.

Content pack and app settings

A content pack combines all the contents that can be updated or published across the web or to various applications, such as datasets, dashboards, reports, Excel workbooks, and so on. As shown in screen 1 of Figure 2-12, there are a few options for handling content packs in Power BI:

- **Publish Content Packs and Apps to the Entire Organization** Enable this feature to allow users to publish content packs to the entire organization.
- **Create Template Organizational Content Packs and Apps** Enable this feature to allow users to create template content packs that use datasets built on one data source in Power BI Desktop.
- **Push Apps to End Users** Enable this feature to allow users to share apps directly with others, without needing to install them from AppSource.

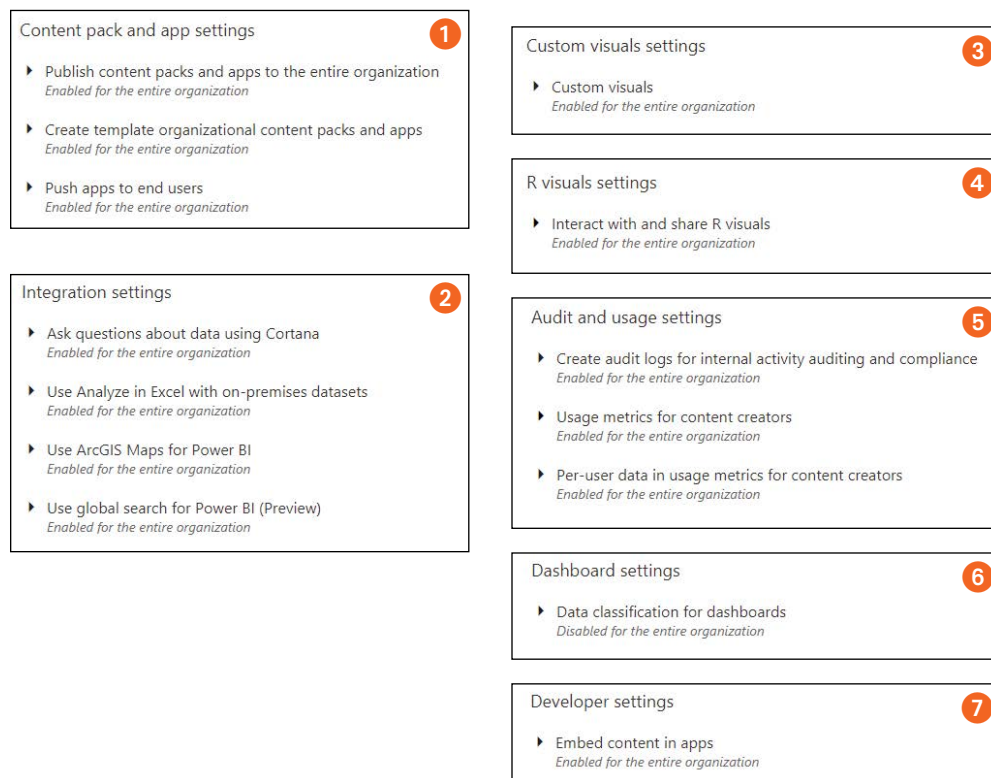


FIGURE 2-12 Additional tenant settings.

Integration settings

The Power BI Admin Portal integration settings give users different options to integrate other tools into Power BI, such as Cortana, Excel, ArcGIS, and Azure to search, ask questions, and analyze data in Power BI reports. As shown in screen 2 in Figure 2-12, these options are as follows:

- **Ask Questions About Data Using Cortana** Enable this feature to allow users to use Cortana to ask specific questions about their data.
- **Use Analyze in Excel with On-Premises Datasets** Enable this feature to allow users to use Excel to view and interact with on-premises Power BI datasets.
- **User ArcGIS Maps for Power BI** Enable this setting to allow users to use the ArcGIS Maps for Power BI visualization provided by Esri.
- **Use Global Search for Power BI** Enable this feature to allow users to employ Global Search for Power BI.



IMPORTANT These settings apply to the entire organization. They cannot be limited to specific individuals or groups.

Custom visuals settings

This setting (see screen 3 in Figure 2-12) enables you to use custom visuals available for Power BI.

R visual settings

As shown in screen 4 in Figure 2-12, there is only one setting in this group: Interact with and Share R Visuals. You can enable this setting to allow users to interact and share visuals created using R scripts. R scripts, commonly called R visuals, can present advanced data shaping and analytics such as forecasting through the use of R.



IMPORTANT This setting applies to the entire organization. It cannot be limited to specific individuals or groups.

Audit and usage settings

This option provides details for usage metrics for parameters such as compliance, per user data, report viewing, and internal activity feature. As shown in screen 5 of Figure 2-12, the options are as follows:

- **Create Audit Logs for Internal Activity Auditing and Compliance** Enable this feature to allow users to monitor actions performed by other users in Power BI.



IMPORTANT This setting must be enabled for audit log entries to be tracked. The setting applies to the entire organization and cannot be limited to specific individuals or groups.

- **Usage Metrics for Content Creators** Enable this feature to allow users to view usage metrics for dashboards and reports they created.
- **Per-User Data in Usage Metrics for Content Creators** Enable this feature if you want usage metrics for content creators to expose the names and email addresses of users who access their content.

Dashboard settings

As shown in screen 6 in Figure 2-12, there is only one setting in this group: Data Classification for Dashboards. You can enable this setting to allow users to tag dashboards to classify the data they contain.



IMPORTANT This setting applies to the entire organization. It cannot be limited to specific individuals or groups.

Developer settings

As shown in screen 7 in Figure 2-12, there is only one setting in this group: Embed Content in Apps. You can enable this setting to allow users to embed Power BI dashboards and reports in various SaaS applications. This setting can be enabled across the organization or limited to specific groups.



IMPORTANT If we disable this feature, users cannot use REST APIs to embed Power BI content in applications.

Capacity Settings

Capacity is the heart of Power BI Premium and Power BI Embedded. It refers to the resources reserved for users to perform functions in Power BI, such as publishing dashboards, reports, and datasets.

You manage capacity using the Capacity Settings page in the Power BI Admin Portal. As shown in Figure 2-13, you can adjust capacity for Power BI Premium and Power BI Embedded. The Power BI Premium tab enables you to add capacity. For Power BI Embedded, capacity admins are assigned from the Azure Portal, which you can access from the Power BI Embedded tab.

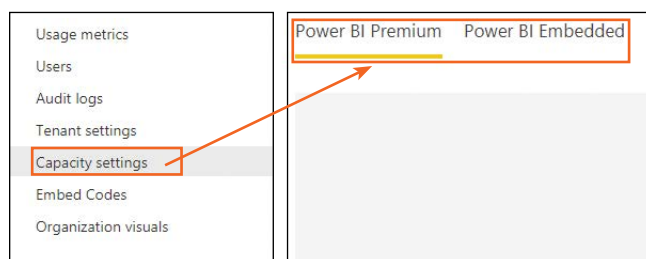


FIGURE 2-13 Capacity Settings in the Power BI Admin Portal.

Embed Codes

As an administrator, you can view, add, or delete the embed codes generated for your Power BI tenant. (See Figure 2-14.) (Embed codes are codes for Power BI reports that can be shared across different applications.)

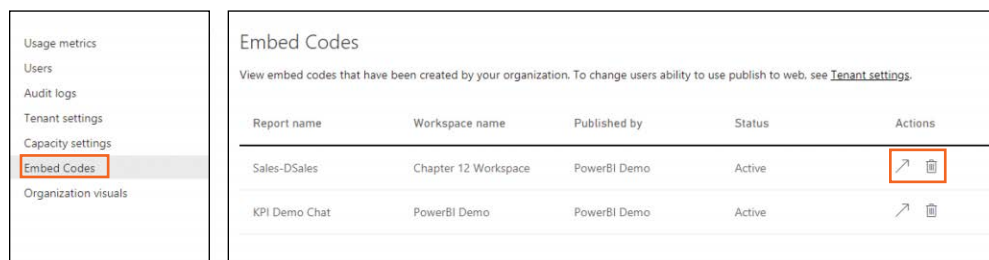


FIGURE 2-14 Embed codes generated by the Power BI tenant.

Organization Visuals

The Organization Visuals page in the Power BI Admin Portal enables users to upload custom visualizations for use by other members of the organization in their own dashboards and reports. It also shows all custom visualizations currently available for this type of use.

To add a custom visualization to Power BI, follow these steps:

1. Click the **Organization Visuals** link in the left pane of the Power BI Admin Portal. (See screen 1 in Figure 2-15.)
2. In the Organization Visuals page, click the **Add a Custom Visual** link. (See screen 2 in Figure 2-15.)
3. The Add Custom Visual dialog box opens. (See screen 3 in Figure 2-15.) Click the **Browse** button next to the **Choose a .pbviz File** box and locate and select the file containing the visualization you want to upload.



IMPORTANT Only versioned API custom visuals are supported. Before uploading a custom visual, be sure it meets your organization's security and privacy standards.

4. **Type a descriptive name for the custom visualization in the Name Your Custom Visuals box.**
5. Click the **Upload** link in the **Icon** section to upload the file that contains the icon you want to use to represent your custom visualization in the Power BI Desktop UI.
6. Type a brief description of the custom visualization in the Description box. Then click **Add**.

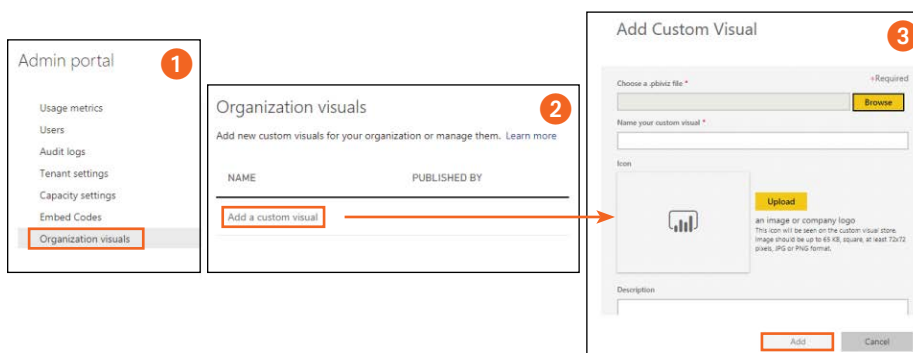


FIGURE 2-15 Adding custom visualizations for use across the organization.

If you need to delete a custom visualization, select it in the list in the Organizational Visuals page, and then click the **Trash Bin** icon.



IMPORTANT Deletion is irreversible. After you delete a visualization, it will no longer be rendered in existing reports. Even if you upload the same visualization again, it will not replace the previous one that was deleted. Users will have to import the new visual and use it to replace the old version in their reports.

Power BI data models and report development

Publishing accurate and insightful Power BI reports involves preparing the data model for the Power BI service or Desktop platform, performing basic operations on the data model, and generating Power BI reports.

Prepare a data model using Power BI service

Data modeling involves streamlining content and shaping data to generate more insightful Power BI reports. Fortunately, Power BI makes this task easy. This section shows you how to prepare a data model in Power BI service using a sample Excel file provided by Microsoft called Sales and Marketing Sample.

You can download the file here: <http://go.microsoft.com/fwlink/?LinkId=529785>.

To prepare a data model in Power BI service, follow these steps:

1. Log in to Power BI service.
2. Click the **Get Data** option in the Power BI service portal. (See screen 1 in Figure 2-16.)
3. In the Import or Connect to Data section, under Files, click the **Get** button. (See screen 2 in Figure 2-16.)
4. Select **Local File**. (See screen 3 in Figure 2-16.)
5. Click the **Import** button in the Import Excel Data into Power BI section. (See screen 4 in Figure 2-16.) A progress bar shows the status of the import operation. A notification will appear to indicate when the operation is complete.

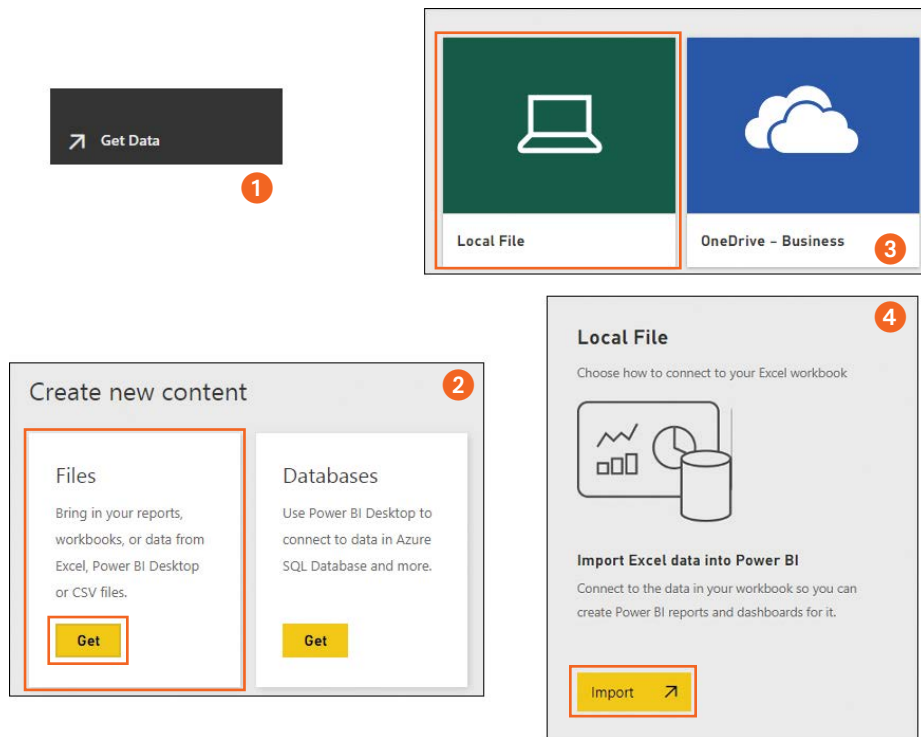


FIGURE 2-16 Preparing a data model in Power BI service.

Prepare a data model in Power BI Desktop

To prepare a data model in Power BI Desktop, follow these steps:

1. Click the **File** menu, select **Import**, and choose **Excel Workbook Contents**. (See screen 1 of Figure 2-17.)
2. Locate and select the **Sales and Marketing Excel** file to import it into Power BI Desktop.
3. An Import Excel Workbook Contents dialog box opens. Click the **Start** button. (See screen 2 of Figure 2-17.) A progress bar shows the status of the import operation.
4. When the operation is complete, a dialog box opens with a **Migration Completed** message. (See screen 3 in Figure 2-17.) Click the **Close** button. Power BI lists the data model—in this case, the contents of the Excel workbook—in a special Data view. (See screen 4 in Figure 2-17.)

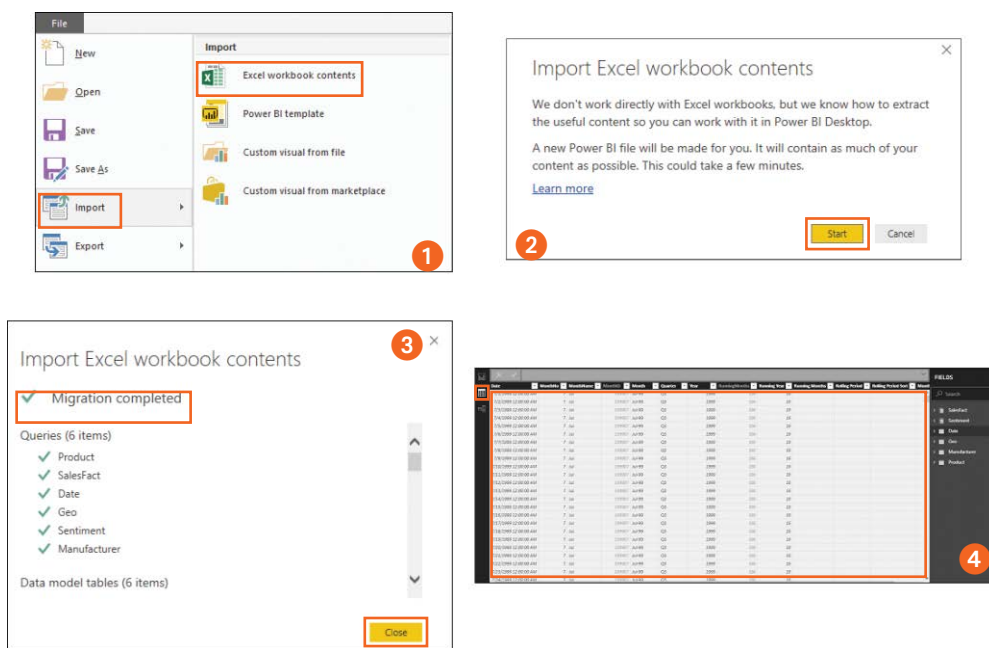


FIGURE 2-17 Creating a data model in Power BI Desktop.

Perform basic data-modeling operations

There are lots of types of data-modelling operations. Some of the most commonly used operations are renaming columns and changing the data type, covered here.

Rename a column

You can rename a column in the Power BI Data view in several different ways. (You can access Data view after data has been loaded into the model.)

- Double-click the column name and type a new name
- Right-click the column name, choose **Rename**, and type a new name. (See image 1 in Figure 2-18.)

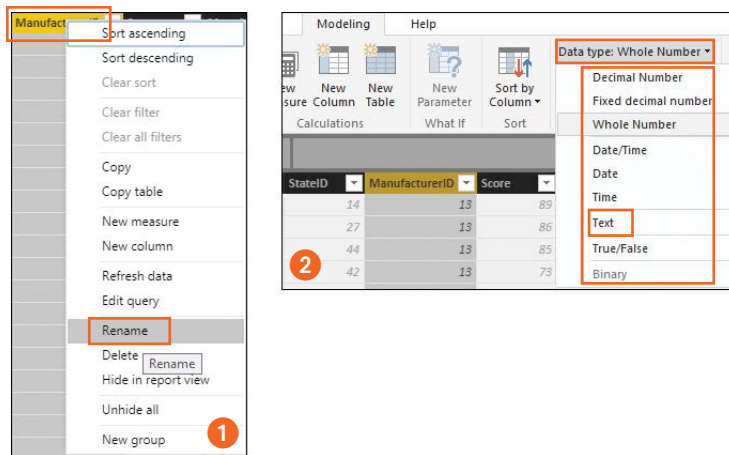


FIGURE 2-18 Renaming a column and changing the data type in Power BI.

Change the data type

There are several data types, available from both Data view and Report view:

- Decimal number
- Fixed decimal number
- Whole number
- Date/Time
- Date
- Time
- Text
- True/False
- Binary

These data types are discussed in more detail in upcoming chapters.

To change the data type of a column, follow these steps:

1. In the Power BI Data view, select the column whose data type you want to change—in this case, the **ManufacturerID** column.
2. Click the **Data Type** tab and choose the desired data type from the menu that appears—for example, **Text**. (See image 2 in Figure 2-18.)

Report development in Power BI

The report-development process depends on which version of Power BI you use: Power BI service or Power BI Desktop.

Prepare reports in Power BI service

Before you can prepare a report in Power BI service, it helps to know how to navigate Power BI service. As shown in Figure 2-19, the Power BI service screen contains two navigation tools: the navigation pane (marked with the numeral 1 on the left side of the screen) and the navigation bar (marked with the numeral 2 along the top).

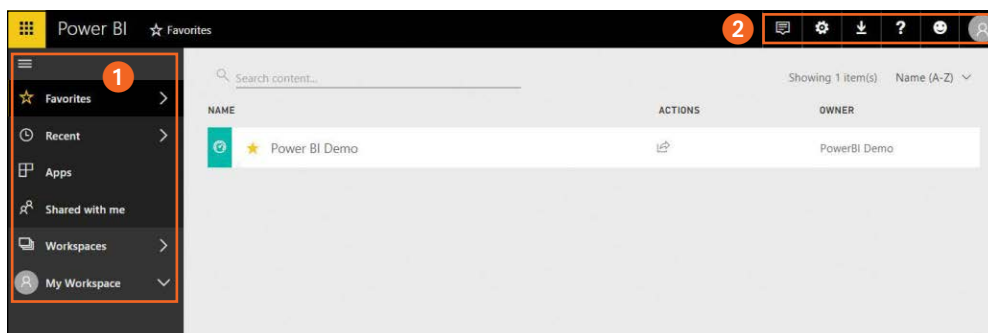


FIGURE 2-19 *Navigational tools in Power BI service.*

The navigation pane contains various options for users to access various reports. These options are as follows:

- **Favorites** Click this option to access reports you've marked as favorites. (You'll learn how to mark a report as a favorite later.)
- **Recent** Click this option to access reports you've opened most recently, in reverse-chronological order.
- **Apps** Click this option to access the entire collection of reports and dashboards.
- **Shared with Me** To access reports that other users have shared with you, click this option.
- **Workspaces** To access workspaces to which you have access, such as a group workspace or a workspace someone else has shared with you, click this option. (You'll learn more about Power BI service workspaces later in this chapter.)
- **My Workspace** Click this to access your own Power BI workspace, from which you can view your dashboards and create and publish your own reports. (Note that the precise name of this option differs depending on what workspaces you have available to you.)

As for the navigation bar, in the top-right corner of the screen, it contains the following icons (from left to right):

- **Notification** Click this icon to view notifications—for example, when one of your reports is updated, when someone shares a report with you, and so on. (If you use the Pro version, you will also receive regular notifications about the number of days remaining on your subscription.)
- **Settings** Click this icon to view statistics on personal storage used, to manage gateways for connecting on-premises reports to Power BI dashboards, create and view content packs, and more.
- **Download** Click this icon to download and save reports available across the Power BI platform.
- **Help and Support** Click this icon to access various help and support resources online, including a help library and user community. (Note that there is a separate help and support section for Power BI developers.)
- **Community Feedback** Click this icon to submit ideas to improve the overall Power BI experience or to report issues.

Power BI Desktop

As shown in Figure 2-20, Power BI Desktop has a more feature-rich main screen than the Online version.

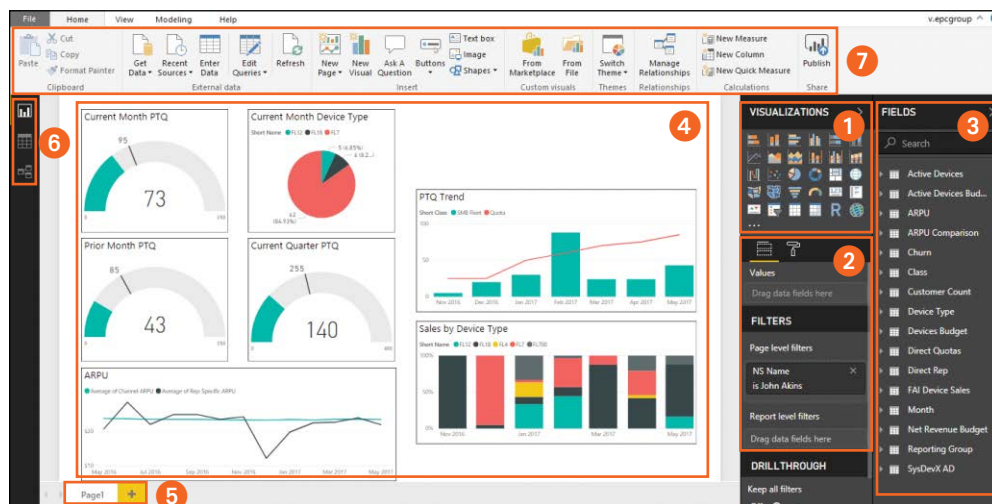


FIGURE 2-20 Navigating Power BI Desktop.

This screen contains the following tabs and features:

- **Visualizations pane** This pane allows easy access to several visualizations, including bar charts, line charts, pie charts, statistical graphs, and more to represent your data in reports. (See numeral 1 in Figure 2-20.)
- **Formatting pane** This pane enables you to apply filters to your data to alter the selected visualization, including report-level filters, page-level filters, and drill-through filters. This pane also offers access to tools to change the appearance of reports such as colors, borders, size, and more. (See numeral 2 in Figure 2-20.)
- **Fields pane** This pane includes the active datasets available for generating reports. To add a dataset to your report, simply select its checkbox in the Fields pane. (See numeral 3 in Figure 2-20.)
- **Report pane** This pane displays visualizations selected by the user from various reports. (See numeral 4 in Figure 2-20.)
- **Pages** Some Power BI reports contain multiple pages. To add a page, you click the plus button at the bottom of the screen. You can also use these controls to display a different page. (See numeral 5 in Figure 2-20.)
- **Display tabs** You can click the Report, Data, and Relationship tabs, marked by numeral 6 in Figure 2-20 and shown again in Figure 2-21, to switch among these three views on the main Power BI Desktop screen. Figure 2-20 shows the Report view. The area marked by the numeral 1 in Figure 2-21 shows a snippet of the Data view. (For a better look at this view, refer to image 4 in Figure 2-17.) Finally, the area marked by the numeral 2 in Figure 2-21 shows the Relationship view.
- **Menu bar** The menu bar allows access to menus for editing templates, opening new files, changing page settings, managing report modeling, accessing help, and more. (See numeral 7 in Figure 2-20.)

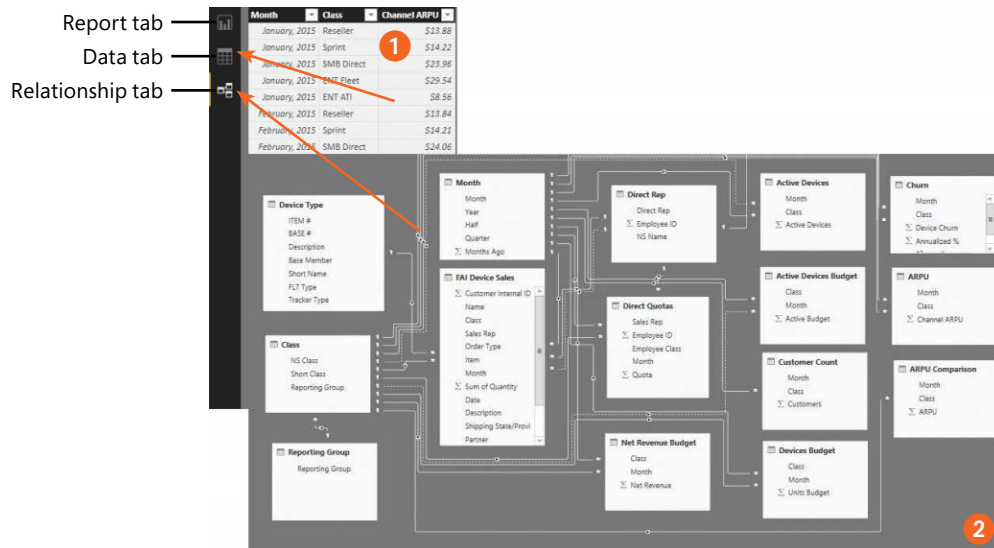


FIGURE 2-21 Use the Display tabs to switch between Report, Data, and Relationship views.

Relationship view

The Relationship view shows how different pieces of data relate to each other. Power BI automatically detects the relationships between datasets and displays them on reports and visualizations. Alternatively, users can establish relationships manually.

Power BI service workspace overview

As shown in Figure 2-22, the Power BI service workspace includes four key screens for analyzing data and creating reports. You access these screens by clicking the corresponding links in the top-left corner of the workspace.

- Dashboards
- Reports
- Workbooks
- Datasets

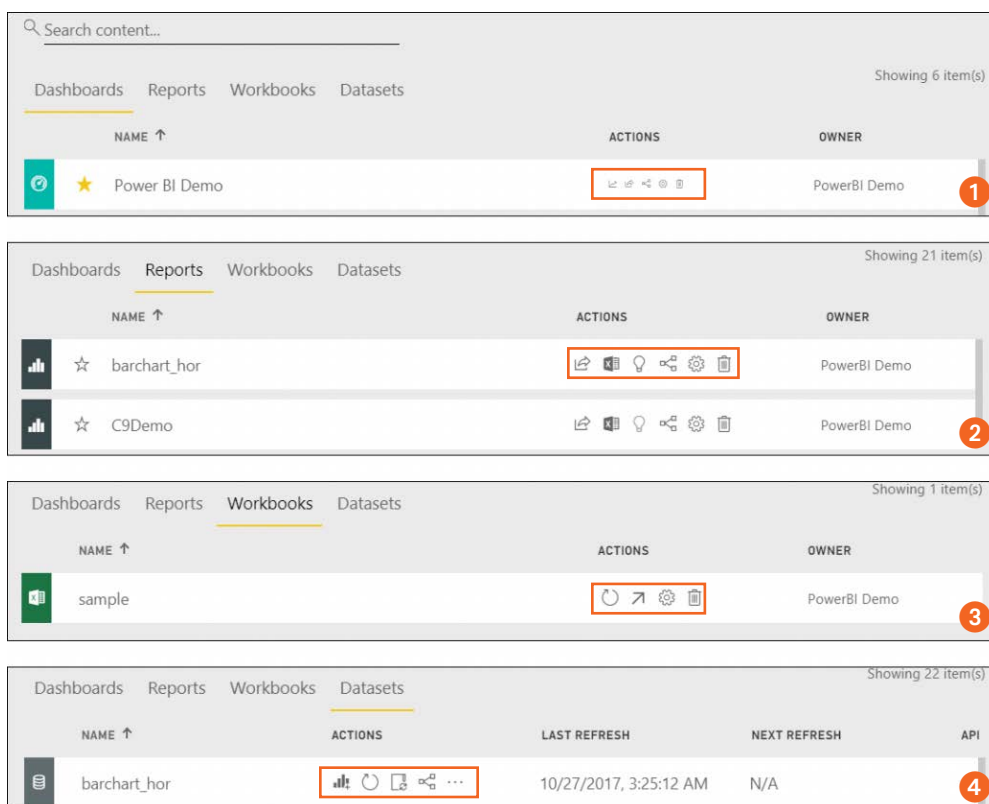


FIGURE 2-22 Components of the Power BI service workspace.

Dashboards

The Dashboard screen shows visualizations of data, such as charts and graphs. Users can customize the dashboard by pinning the visualizations they need most often. There are also buttons that allow the user to perform different actions related to the selected dashboard. (See screen 1 in Figure 2-22.)

- **Usage Metrics** Click this button to view usage metrics—including how many people have viewed the dashboard today and the number of unique users for days prior—to gauge user engagement.
- **Share** Click this button to share the dashboard with others.

- **View Related** When you click this button, Power BI checks for related content in the dataset servers. (Note that you can also schedule Power BI to fetch this data automatically.)
- **Settings** Click this to manage Q&A settings and the dashboard tile flow. You can also use this option to change the name of the dashboard.
- **Delete** Click this button to permanently delete the dashboard.

Reports

The Reports screen contains a list of reports created by the user. As with the Dashboard screen, this screen features buttons that allow the user to perform different actions related to the selected report. (See screen 2 in Figure 2-22.) These include Usage Metrics, View Related, Settings, and Delete buttons. The Reports screen also includes the following:

- **Analysis in Excel** Click this button (second from the left) to download the report into Excel. (Note that to use this feature, a one-time update is required for Excel libraries.)
- **Quick Insights** Click this button (third from the left) to view a list of all visualizations available in a report to quickly glean key insights. You can then pin any critical visualizations to the dashboard.

Workbooks

You can upload Excel files into Power BI. When you do they appear in the Workbooks screen. As with the other screens, the Workbooks screen contains buttons you can use to perform different actions on the selected workbook. (See screen 3 in Figure 2-22.)

- **Refresh** Click this button to refresh the data in the selected Excel file.
- **Replace** To replace the selected Excel file with a different one, click this button.
- **Settings** Click this button to access settings that enable you to change the name of the Excel file and see where it is stored on the local server.
- **Delete** Click this button to delete the Excel file.

Datasets

This screen lists the datasets for all reports created by the user. As with the other screens, the Datasets screen contains buttons you can use to perform different actions on the selected dataset. (See screen 4 in Figure 2-22.)

- **Create Report** Click this button to create a new report using the selected dataset.
- **Refresh** Click this button to refresh the dataset.
- **Schedule Refresh** Click this button to schedule a refresh for the dataset.
- **View Related** Click this button to find related datasets.

Edit options in Power BI service

The menu bar in Power BI service (see Figure 2-23) allows access to menus and buttons with several key options. These include the following:

- **File** This menu includes the following commands. (These are quite self-explanatory.)
 - Save As
 - Print
 - Publish to Web
 - Embed in SharePoint Online
 - Export to PowerPoint
 - Download Report
- **View** You can use this menu to access the following view options for Power BI reports:
 - Fit to page
 - Fit to width
 - Actual size
- **Edit Report** Access editing tools to edit your report here.
- **Explore** This provides options for navigating through datasets.
- **Refresh** Click this button to refresh the selected report.

- **Pin Live Page** Click this option to pin the current page in the selected report to your dashboard.
- **View Related** Click this option to view the Related Content pane, which shows you how your Power BI service content—that is, your dashboards, reports, and datasets—are interconnected.
- **Subscribe** Click this button to receive email notifications about changes to the current Power BI report page. (Emails are limited to one per day—assuming the report has been updated.)
- **Generate QR Code** Click this button to create a report-specific QR code that can be accessed from a mobile device.
- **Analyze in Excel** Click this button to open the report in tabular form in Microsoft Excel.



FIGURE 2-23 *The Power BI service menu bar.*

Skills review

In this chapter, you learned about:

- Power BI architecture
- Data processing in Power BI
- How Power BI consumes the OLAP cube for report generation
- The Power BI Admin Portal
- A general overview of data models and report development



Practice tasks

This section provides simple case studies for you to study and solve using information from this chapter. This section also contains a series of practice questions for you to answer.

Case study 1

Consider a large organization that sells shoes on a global scale. They market thousands of shoe styles in specific regions around the world. Regional business units provide the corporate office with real-time reports of daily sales as well as regional marketing reports on a daily basis.

The data used to generate these reports is stored in Azure SQL databases as well as several on-premises non-SQL data sources. This makes it difficult to summarize this information, to collaborate with it, and to obtain specific types of reports used by corporate teams. Employees are left unable to answer analytical questions such as the following:

- How many regions are involved in the business?
- How can we as an organization quickly adapt to the changing needs of our customers?
- How can targeted sales and discounts help specific regions achieve their goals?
- What are the purchasing rates per individual according to region?
- How do targeted marketing campaigns affect sales in specific regions?

Power BI provides the organization with a way to answer these questions by capturing data and conveying it in a more refined way. Users can prepare a single dataset from sourced information and analyze it to glean powerful insights.

Case study 2

Suppose you have already created a Power BI report online, which is available for consumption among multiple users. To gain a better understanding of the various dynamics of report editing, complete the following tasks:

- Generate a SharePoint embedded URL.
- Export a specific report page from Power BI service.
- Generate an embedded URL in Power BI.
- Obtain a QR code for a report for sharing with other users.

Practice questions

1. Outline the ETL process as it relates to Power BI.
2. What phases are involved in Power BI report development?
3. How do you open the Power BI Admin Portal?
4. What process is involved in creating an OLAP cube from raw data?
5. How do you disable the sharing feature in any Power BI report?
6. Name the main features of Power BI Admin Portal.
7. What are embed codes?
8. How do you add a custom visualization to Power BI?
9. What are the key tenant settings available in the Power BI Admin Portal?

This page intentionally left blank

Visuals in Power BI

3

This chapter provides an overview of Power BI visuals, or visualizations, available for users to represent their data and for related analysis.

This chapter discusses out-of-the-box (OOTB) visuals from Power BI, and in what scenarios each type of visual works best. It also covers third-party visuals that can be used in Power BI—including how to download and import them into the application.

This chapter also talks about how to drill through data in Power BI and about filters available in Power BI. Filters enable you to choose precisely what data should be depicted by Power BI visuals.

In this chapter:

- Overview of visuals in Power BI
- Power BI out-of-the-box (OOTB) visuals
- Third-party visuals in Power BI
- Drill through visuals in Power BI
- Apply filters in Power BI reports